

Appendix

This appendix is the canonical ledger for the CTCC cost model that backs the Methods and case studies in the main text. Each section below lists variables, units, equations, and conditional logic for one module so the mapping from engineering inputs to dollar totals is auditable. All cost and benefit symbols (C , B) denote present values unless explicitly marked ,nominal.

The appendix is organized into five top-level sections. **Utilities** defines shared building blocks (weighted miles, discount rates, AFUDC and rate base, system constraints). **Costs** groups modules into four buckets (hard, soft, risk, emissions). **Benefits** groups modules into remedial and enabling buckets plus displacement reporting. **Reporting Framework** assembles bucket totals, net benefit, incrementals, and BCR perspectives. **Revenue** defines the utility–ratepayer transfer.

A Utilities

This section defines shared mathematical building blocks that multiple cost and benefit modules depend on. Each utility is defined once and referenced throughout the cost and benefit modules in subsequent sections: weighted miles (terrain-adjusted distance), financial parameters (discount rates and capital structure), AFUDC and rate base, and system constraints and capacity relief.

A.1 Weighted Miles

This subsection defines weighted miles, the difficulty-adjusted line miles used for build costs and other modules that use terrain-adjusted distance (consumed by build costs §B.1.1, ROW capital §B.1.2, and environmental mitigation §B.1.3). The terrain set is the set of terrain types: Forested, Scrubbed Flat, Wetland, Farmland, Desert/Barren, Urban, Rolling Hills, Mountain, Subsea. Only terrains with $M_{\text{terrain}} > 0$ are used in sums below and elsewhere in this appendix.

Variables

Variable	Meaning / units	Notes
M_{terrain}	Miles of a specific terrain type, (miles)	Terrain set as defined in this subsection. Only terrains with $M_{\text{terrain}} > 0$ are used in sums.
λ_{terrain}	Cost multiplier by terrain type, (dimensionless, ≥ 1)	Same terrain list as M_{terrain} .
M_{total}	Total (raw) miles, (miles)	Sum of M_{terrain} over all terrains.
$M_{\text{weighted,terrain}}$	Weighted miles for that terrain type, (miles)	$M_{\text{weighted,terrain}} = M_{\text{terrain}} \times \lambda_{\text{terrain}}$ for that terrain.
$M_{\text{weighted,total}}$	Total weighted miles across all terrain types, (miles)	Used in Build costs (§B.1.1) for conductor/structure/converter variable costs.
λ_{average}	Average terrain multiplier for the project, (dimensionless)	$\lambda_{\text{average}} = M_{\text{weighted,total}}/M_{\text{total}}$ when $M_{\text{total}} > 0$.

Equations

1. Weighted miles for each terrain type

$$M_{\text{weighted,terrain}} = M_{\text{terrain}} \times \lambda_{\text{terrain}} \quad \text{for that terrain type.}$$

2. Total weighted miles

$$M_{\text{weighted,total}} = \sum_{\text{terrain}} M_{\text{weighted,terrain}}.$$

3. Total (raw) miles

$$M_{\text{total}} = \sum_{\text{terrain}} M_{\text{terrain}}.$$

4. Average terrain multiplier

$$\lambda_{\text{average}} = \frac{M_{\text{weighted,total}}}{M_{\text{total}}} \quad (M_{\text{total}} > 0).$$

A.2 Financial Parameters

This subsection defines discount rates, inflation, and base year used across CTCC. The real WACC ($r_{\text{WACC,real}}$) discounts all market-valued costs and benefits; the social discount rate (r_{social}) discounts emissions externalities (§B.4).

Variables

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Variable	Meaning / units	Notes
π	Annual inflation rate, (decimal)	Used in Fisher link for real WACC.
Y_{base}	Base (reference) year, (year)	Reference year for all present values; PV is in base-year dollars.
$r_{\text{WACC,nom}}$	Nominal weighted average cost of capital, (decimal)	Direct input. Used for AFUDC/utility perspective.
$r_{\text{WACC,real}}$	Real weighted average cost of capital, (decimal)	Used for societal PV of costs and benefits. Calculated via Fisher link.
r_{social}	Social discount rate, (decimal)	Used for externality costs (emissions, expected wildfire cost, expected outage cost).

Equations

1. Real WACC (Fisher link)

$$r_{\text{WACC,real}} = \frac{1 + r_{\text{WACC,nom}}}{1 + \pi} - 1.$$

When each rate is used

- $r_{\text{WACC,nom}}$: Utility/regulatory perspective (e.g. AFUDC rate when applicable).

- $r_{\text{WACC,real}}$: Societal PV of capital, O&M, insurance, ROW rent, energy loss costs, benefits, revenue; delay and congestion/curtailment delay costs.
- r_{social} : Societal PV of externality costs: emissions (§B.4), expected cost of wildfires (§B.3.1), expected cost of outages (§B.3.2).

A.3 AFUDC and Rate Base

This subsection defines allowance for funds used during construction (AFUDC) and rate base. AFUDC compounds nominal spending on construction work in progress (CWIP) to the commercial operation date (COD). Rate base at COD is the sum of capitalized AFUDC-eligible capital: build, ROW capital (acquisition and holding), and environmental mitigation. The nominal WACC from §A.2 is used as the AFUDC rate. Costs not AFUDC-eligible (e.g. delay, O&M, ROW rent, wildfire/outage) do not enter rate base.

Variables

Variable	Meaning / units	Notes
i	Component index, (index)	$i \in \{\text{build, ROW, env mitigation}\}$.
C_i^{nom}	Nominal spending (pre-AFUDC) for component i , (\$)	Same timing as used for AFUDC.
θ_{delay}	Fraction of C_i^{nom} incurred during delay, (dimensionless)	For AFUDC-eligible: $\theta_{\text{delay}} + \theta_{\text{construction}} = 1$.
$\theta_{\text{construction}}$	Fraction of C_i^{nom} incurred during construction, (dimensionless)	For AFUDC-eligible: $\theta_{\text{delay}} + \theta_{\text{construction}} = 1$.
T_{delay}	Delay years, (years)	From project technical details.
$T_{\text{construction}}$	Construction years, (years)	From project technical details.
$n_{\text{construction}}$	Compound period for construction portion, (years)	Years from midpoint of construction to COD; $n_{\text{construction}} = T_{\text{construction}}/2$.
n_{delay}	Compound period for delay portion, (years)	Depends on timing: midpoint of delay to COD, or construction only, or 0; see paragraph 2b.
r_{AFUDC}	AFUDC rate (annual), (decimal)	From §A.2: $r_{\text{AFUDC}} = r_{\text{WACC,nom}}$.
AFUDC_i	AFUDC allowance for component i , (\$)	$C_i^{\text{cap}} - C_i^{\text{nom}}$.
C_i^{cap}	Capitalized cost at COD for component i , (\$)	$C_i^{\text{cap}} = C_i^{\text{nom}} + \text{AFUDC}_i$.
RB_{nominal}	Nominal rate base at COD, (\$)	Sum of C_i^{cap} over build, ROW, env mitigation.

Equations

1. AFUDC rate.

$$r_{\text{AFUDC}} = r_{\text{WACC,nom}} \quad (\text{from §A.2}).$$

2. Capitalization to COD. For each AFUDC-eligible component i , nominal spending is split by timing $(\theta_{\text{delay}}, \theta_{\text{construction}})$. The delay portion (if any) is compounded to COD; the construction portion is compounded from midpoint of construction to COD. Then $C_i^{\text{cap}} = C_i^{\text{nom}} + \text{AFUDC}_i$. The compound periods and formula are given in paragraph 2b.

2b. Timing methodology (compound periods and formula). Spending within each period is treated as occurring at the midpoint of that period for compounding. Let $r = r_{\text{AFUDC}}$.

Construction portion: years from midpoint of construction to COD:

$$n_{\text{construction}} = T_{\text{construction}}/2.$$

Delay portion: compound period n_{delay} depends on timing:

- Active CWIP during delay (e.g. pre-construction work that earns AFUDC): $n_{\text{delay}} = T_{\text{construction}} + T_{\text{delay}}/2$ (midpoint of delay to COD).
- No active work during delay (AFUDC suspended) and $T_{\text{delay}} > 0$: $n_{\text{delay}} = T_{\text{construction}}$ (delay portion compounded only over construction period).
- $T_{\text{delay}} = 0$: $n_{\text{delay}} = 0$ (delay term at face value).

Capitalized cost at COD:

$$C_i^{\text{cap}} = (\theta_{\text{delay}} C_i^{\text{nom}})(1 + r)^{n_{\text{delay}}} + (\theta_{\text{construction}} C_i^{\text{nom}})(1 + r)^{n_{\text{construction}}}, \quad \text{where } r = r_{\text{AFUDC}}.$$

Either term is omitted if the corresponding period is absent. Hence $\text{AFUDC}_i = C_i^{\text{cap}} - C_i^{\text{nom}}$.

3. Rate base at COD.

$$RB_{\text{nominal}} = C_{\text{build}}^{\text{cap}} + C_{\text{ROW}}^{\text{cap}} + C_{\text{env mit}}^{\text{cap}}.$$

Which costs enter rate base. Build costs (§B.1.1), capital ROW (§B.1.2), and environmental mitigation (§B.1.3) are the three components in rate base. Revenue (§E) uses RB_{nominal} to compute revenue from the utility/ratepayer perspective.

A.4 System Constraints and Relief

This subsection defines the system constraints a project addresses and the capacity relief it provides. Effective capacity relief ($\Delta C_{\text{effective}}$) and curtailment-first allocation feed congestion benefit (§C.1.2), curtailment benefit (§C.1.1), delivered energy (§C.2.1), and facilitated emissions (§B.4.2).

Variables

Variable	Meaning / units	Notes
$\Delta C_{\text{effective}}$	Effective capacity relief, (MW)	
ϕ	Flow factor, (dimensionless [0,1])	Greenfield: fraction of nameplate that relieves constraint.
C_{new}	New line capacity, (MW)	Greenfield and reconductoring.
C_{old}	Old line capacity, (MW)	Reconductoring only.
θ	Overlap of binding and curtailment hours, (dimensionless)	0 = no overlap, 1 = full overlap; 0 if only congestion or only curtailment.
$H_{\text{congestion}}$	Hours of congestion per year, (h/year)	Binding hours.
$H_{\text{curtailment}}$	Hours of curtailment per year, (h/year)	
$H_{\text{congestion,overlap}}$	Congestion hours that overlap with curtailment, (h/year)	
$H_{\text{congestion,nonoverlap}}$	Congestion hours that do not overlap with curtailment, (h/year)	
$X_{\text{congestion}}$	MW of congestion (exceedance), (MW)	Average exceedance during binding hours.
$MW_{\text{curtailment}}$	MW of curtailed resources, (MW)	Average curtailed MW during curtailment hours.
$\Delta C_{\text{remaining}}$	Capacity remaining for congestion after curtailment, (MW)	Curtailment is addressed first.
$R_{\text{curtailment,annual}}$	Curtailment relief due to project, (MWh/year)	
$R_{\text{congestion,overlap,annual}}$	Overlap-hours congestion relief, (MWh/year)	
$R_{\text{congestion,nonoverlap,annual}}$	Non-overlap-hours congestion relief, (MWh/year)	
$R_{\text{congestion,annual}}$	Congestion relief due to project, (MWh/year)	Sum of overlap and non-overlap.

Equations

Curtailment is addressed first: curtailment relief (paragraph 4) uses $\Delta C_{\text{effective}}$; the capacity remaining after curtailment, $\Delta C_{\text{remaining}}$ (paragraph 2), is then used for congestion relief (paragraph 5).

1. Effective relief.

$$\Delta C_{\text{effective}} = \begin{cases} \phi \times C_{\text{new}} & \text{if greenfield} \\ C_{\text{new}} - C_{\text{old}} & \text{otherwise} \end{cases}$$

2. Capacity remaining to address congestion (curtailment first).

$$\Delta C_{\text{remaining}} = \max(0, \Delta C_{\text{effective}} - \min(\Delta C_{\text{effective}}, MW_{\text{curtailment}}))$$

3. Overlap of congestion and curtailment hours.

$$\theta = \begin{cases} \min\left(1, \frac{H_{\text{curtailment}}}{H_{\text{congestion}}}\right) & \text{if } H_{\text{curtailment}} > 0 \text{ and } H_{\text{congestion}} > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$H_{\text{congestion,overlap}} = \theta \times H_{\text{congestion}}, \quad H_{\text{congestion,nonoverlap}} = (1 - \theta) \times H_{\text{congestion}}.$$

4. Curtailment relief (MWh/year).

$$R_{\text{curtailment,annual}} = H_{\text{curtailment}} \times \min(\Delta C_{\text{effective}}, MW_{\text{curtailment}}).$$

5. Congestion relief (MWh/year). Total congestion relief is the sum of overlap and non-overlap components:

$$R_{\text{congestion,annual}} = R_{\text{congestion,overlap,annual}} + R_{\text{congestion,nonoverlap,annual}}.$$

Overlap-hours relief (capacity remaining after curtailment, $\Delta C_{\text{remaining}}$):

$$R_{\text{congestion,overlap,annual}} = H_{\text{congestion,overlap}} \times \min(\Delta C_{\text{remaining}}, X_{\text{congestion}}).$$

Non-overlap-hours relief (full $\Delta C_{\text{effective}}$):

$$R_{\text{congestion,nonoverlap,annual}} = H_{\text{congestion,nonoverlap}} \times \min(\Delta C_{\text{effective}}, X_{\text{congestion}}).$$

B Costs

This section defines the four cost buckets: hard costs (C_{hard}), soft costs (C_{soft}), risk costs (C_{risk}), and emissions costs ($C_{\text{emissions}}$). The grand total cost for scenario P is $C^P = C_{\text{hard}}^P + C_{\text{soft}}^P + C_{\text{risk}}^P + C_{\text{emissions}}^P$.

B.1 Hard Costs (C_{hard})

Hard costs are one-time capital expenditures that are AFUDC-eligible and enter rate base at COD: build costs, capital ROW (acquisition and holding), and environmental mitigation (see §A.3). These three components produce the hard cost bucket C_{hard} (§B.1.3).

B.1.1 Build costs

This subsection defines build cost: terrain-adjusted and contingency-applied conductor, structure, and converter costs. Build cost is AFUDC-eligible and enters rate base at COD. Component costs use weighted miles (§A.1); reconductoring excludes structure and converter. Component costs use weighted miles (§A.1); reconductoring excludes structure and converter.

Variable	Meaning / units	Notes
$C_{\text{conductor}}$	Conductor cost (\$)	Not terrain adjusted, no contingency applied.
$C_{\text{structure}}$	Structure cost (\$)	Not terrain adjusted, no contingency applied.
$C_{\text{converter}}$	Converter cost (if DC) (\$)	Not terrain adjusted, no contingency applied.
$C_{\text{adj,conductor}}$	Conductor cost (\$)	Terrain adjusted, no contingency applied.
$C_{\text{adj,structure}}$	Structure cost (\$)	Terrain adjusted, no contingency applied.
$C_{\text{adj,converter}}$	Converter cost (if DC) (\$)	Terrain adjusted, no contingency applied.
$C_{\text{adj,cont,conductor}}$	Conductor cost (\$)	Terrain adjusted, contingency applied.
$C_{\text{adj,cont,structure}}$	Structure cost (\$)	Terrain adjusted, contingency applied.
$C_{\text{adj,cont,converter}}$	Converter cost (if DC) (\$)	Terrain adjusted, contingency applied.
$F_{\text{conductor}}$	Fixed conductor costs (\$)	Conductors have both fixed and variable costs.
$F_{\text{structure}}$	Fixed structure costs (\$)	Currently, structures have no fixed costs in the CTCC.
$F_{\text{converter}}$	Fixed converter costs (\$)	Converters only have fixed costs.
$V_{\text{conductor}}$	Variable conductor costs, (\$/mile)	Miles used are terrain adjusted (§A.1).
$V_{\text{structure}}$	Variable structure costs, (\$/mile)	Miles used are terrain adjusted.
$V_{\text{converter}}$	Variable converter costs, (\$/mile)	Converters currently have no variable costs — only fixed.
$M_{\text{weighted,total}}$	Total weighted miles, (miles)	From §A.1.
$r_{\text{contingency,conductor}}$	Contingency multiplier conductors, (dimensionless [0,1])	
$r_{\text{contingency,structure}}$	Contingency multiplier structures, (dimensionless [0,1])	
$r_{\text{contingency,converter}}$	Contingency multiplier converter, (dimensionless [0,1], if DC)	
C_{build}	Total build cost, nominal (\$)	Terrain adjusted, contingency applied.
C_{build}	Total build cost (PV) (\$)	Societal perspective; spending spread over construction and discounted at $r_{\text{WACC,real}}$.
T_{delay}	Delay years, (years)	From project technical details.
$T_{\text{construction}}$	Construction years, (years)	From project technical details.
t_{start}	Construction start year, (year index)	$t_{\text{start}} = T_{\text{delay}} + 1$ (1-based).
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Societal discount rate (§A.2).
ξ_{DC}	Binary for DC projects, (0 or 1)	1 if DC, 0 if AC.
$\xi_{\text{reconductoring}}$	Binary for reconductoring, (0 or 1)	0 if reconductoring, 1 if not.

Variables

Equations

1. Component costs (raw miles).

$$C_{\text{conductor}} = V_{\text{conductor}} \cdot M_{\text{total}} + F_{\text{conductor}}, \quad C_{\text{structure}} = V_{\text{structure}} \cdot M_{\text{total}} + F_{\text{structure}}, \quad C_{\text{converter}} = V_{\text{converter}} \cdot M_{\text{total}} + F_{\text{converter}}.$$

In CTCC, $F_{\text{structure}} = 0$ and $V_{\text{converter}} = 0$.

2. Terrain-adjusted component costs.

$$C_{\text{adj,conductor}} = V_{\text{conductor}} \cdot M_{\text{weighted,total}} + F_{\text{conductor}}, \quad C_{\text{adj,structure}} = V_{\text{structure}} \cdot M_{\text{weighted,total}}, \quad C_{\text{adj,converter}} = F_{\text{converter}}.$$

3. Contingency-applied costs.

$$C_{\text{adj,cont,conductor}} = (1 + r_{\text{contingency,conductor}}) \cdot C_{\text{adj,conductor}}, \quad C_{\text{adj,cont,structure}} = (1 + r_{\text{contingency,structure}}) \cdot C_{\text{adj,structure}},$$

$$C_{\text{adj,cont,converter}} = (1 + r_{\text{contingency,converter}}) \cdot C_{\text{adj,converter}}.$$

4. Total build cost.

$$\text{II} \quad C_{\text{build}} = C_{\text{adj,cont,conductor}} + \xi_{\text{reconductoring}} \cdot C_{\text{adj,cont,structure}} + \xi_{\text{reconductoring}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}.$$

Regulatory perspective (AFUDC). Build cost is AFUDC-eligible and is capitalized to COD using the same logic as other capital costs (see §A.3). The nominal spending C_{build} is split by timing (θ_{delay} , $\theta_{\text{construction}}$) and compounded to COD.

Societal perspective (present value). Build spending is spread evenly over $T_{\text{construction}}$ starting at $t_{\text{start}} = T_{\text{delay}} + 1$ and discounted at real WACC. For $T_{\text{construction}} > 0$:

$$C_{\text{build,annual}} = \frac{C_{\text{build}}}{T_{\text{construction}}}, \quad C_{\text{build}} = \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{C_{\text{build,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

If $T_{\text{construction}} = 0$, the full cost is one-time at t_{start} : $C_{\text{build}} = C_{\text{build}} / (1 + r_{\text{WACC,real}})^{t_{\text{start}}}$.

B.1.2 Capital ROW costs (acquisition, holding)

This subsection defines capital ROW costs: one-time acquisition and annual holding (option fee) over the delay period. These costs are AFUDC-eligible and enter rate base at COD. ROW rent is not a capital cost; it is an operational cost (§B.2.3). ROW here refers only to the linear corridor (miles $\times$ width by zone). Converter station acreage is not modeled.

Variable	Meaning / units	Notes
zone	Zone index, (index)	ROW split into zones (1–15). Use only zones where $M_{\text{zone}} > 0$.
M_{zone}	Miles in zone, (miles)	
W_{ROW}	ROW width, (ft)	Contingent on project category.
A_{zone}	Area of ROW of zone, (acres)	
$p_{\text{acquisition,zone}}$	Acquisition cost per acre, zone, (\$/acre)	One-time.
$p_{\text{hold,zone}}$	Holding (option fee) per acre per year, zone, (\$/acre/year)	
$p_{\text{rent,zone}}$	Rent per acre per year, zone, (\$/acre/year)	Operational only (§B.2.3).
$C_{\text{acquisition}}$	Total one-time acquisition cost (\$)	From zone sum.
$C_{\text{hold,annual}}$	Total annual holding cost (\$/year)	Permitting/delay years.
T_{delay}	Delay years, (years)	Used for holding.
$C_{\text{hold,total}}$	Total holding cost over delay (\$)	Option fee over full delay period.
$\xi_{\text{acquisition}}$	Binary: acquisition counts toward capital ROW, (0 or 1)	1 if acquisition applies, 0 if not (e.g. lease). Set by agreement type.
ξ_{holding}	Binary: holding counts toward capital ROW, (0 or 1)	1 if holding applies, 0 if not. Set by agreement type.
$C_{\text{acquisition}}$	PV of acquisition (\$)	Societal; discounted at $r_{\text{WACC,real}}$.
C_{hold}	PV of holding (\$)	Societal; discounted at $r_{\text{WACC,real}}$.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Societal discount rate (§A.2).
$C_{\text{ROW,capital,nominal}}$	Total ROW capital costs, nominal (\$)	
$C_{\text{ROW,capital}}$	Total ROW capital costs (PV) (\$)	Societal perspective.

Agreement type	Acquisition	Holding
New Permanent Easement	Yes ($\xi_{\text{acquisition}} = 1$)	Yes ($\xi_{\text{holding}} = 1$)
Simple Fee	Yes ($\xi_{\text{acquisition}} = 1$)	Yes ($\xi_{\text{holding}} = 1$)
Existing	No ($\xi_{\text{acquisition}} = 0$)	No ($\xi_{\text{holding}} = 0$)
Lease/License		
Federal Hybrid	Yes ($\xi_{\text{acquisition}} = 1$)	Yes ($\xi_{\text{holding}} = 1$)

Variables

Equations

1. Area of ROW per zone.

$$A_{\text{zone}} = \frac{M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

2. Total acquisition cost.

$$C_{\text{acquisition}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{acquisition,zone}}.$$

3. Annual holding cost.

$$C_{\text{hold,annual}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{hold,zone}}.$$

4. Total holding cost over delay.

$$C_{\text{hold,total}} = C_{\text{hold,annual}} \times T_{\text{delay}}.$$

5. Total nominal ROW capital cost.

$$C_{\text{ROW,capital,nominal}} = \xi_{\text{acquisition}} \cdot C_{\text{acquisition}} + \xi_{\text{holding}} \cdot C_{\text{hold,total}}.$$

Regulatory perspective (AFUDC). ROW acquisition and holding are AFUDC-eligible and are capitalized to COD using the same logic as other capital costs (see §A.3). A timing pattern is applied: acquisition is one-time; holding is incurred over the delay period.

Societal perspective (present value). Discount at real WACC:

$$C_{\text{acquisition}} = \frac{C_{\text{acquisition}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}}}}, \quad C_{\text{hold}} = \sum_{t=1}^{T_{\text{delay}}} \frac{C_{\text{hold,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

Total real (present) value of ROW capital costs:

$$C_{\text{ROW,capital}} = \xi_{\text{acquisition}} \cdot C_{\text{acquisition}} + \xi_{\text{holding}} \cdot C_{\text{hold}}.$$

B.1.3 Environmental mitigation

This subsection defines environmental mitigation costs: base costs (construction/restoration per effective acre by terrain and construction type) and credit costs (wetland and habitat off-site mitigation). The **habitat set** is the set of habitat terrain types: Forested, Scrubbed Flat, Desert/Barren, Rolling Hills, Mountain. The **wetland set** is the set of wetland terrain types: Wetland. Only habitat terrains with $M_{\text{terrain}} > 0$ and wetland terrains with $M_{\text{terrain}} > 0$ are used. For reconductoring projects, wetland and habitat credits are set to zero. Wetland credits use a single per-acre rate. Habitat credits use per-terrain rates.

Variable	Meaning / units	Notes
M_{terrain}	Miles of a specific terrain type, (miles)	Terrain set as in §A.1. Only terrains with $M_{\text{terrain}} > 0$ used.
W_{ROW}	ROW width, (ft)	Contingent on project category.
A_{terrain}	Base ROW area in terrain, (acres)	$A_{\text{terrain}} = (M_{\text{terrain}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}) / 43\,560 \text{ (ft}^2\text{/acre)}$.
$u_{\text{mitigation}}$	Mitigation uplift factor, (dimensionless)	Dimensionless (e.g. 1.25). Construction width beyond ROW (TCE).
$A_{\text{terrain}}^{\text{effective}}$	Effective acres for mitigation in terrain, (acres)	$A_{\text{terrain}}^{\text{effective}} = A_{\text{terrain}} \times u_{\text{mitigation}}$.
CT	Construction type, (index)	From project category (overhead, underground direct buried, tunnel, subsea).
$c_{\text{base,peracre}}(\text{terrain}, CT)$	Base mitigation cost per acre, (\$/acre)	Depends on terrain and construction type.
$C_{\text{base,total}}$	Total base mitigation/restoration cost (\$)	Sum over terrains ($M_{\text{terrain}} > 0$) of $c_{\text{base,peracre}} \times A_{\text{terrain}}^{\text{effective}}$.
$A_{\text{wetland}}^{\text{effective impact}}$	Wetland impact acres (effective), (acres)	Wetland terrain only: $A_{\text{wetland}} \times u_{\text{mitigation}}$. Use only when $M_{\text{terrain}} > 0$ for wetland.
$A_{\text{habitat}}^{\text{effective impact}}$	Habitat impact acres (effective), total, (acres)	Sum over habitat terrains $\times u_{\text{mitigation}}$. Habitat = forested, scrubbed flat, desert barren, rolling hills, mountain. Use only when $M_{\text{terrain}} > 0$ for habitat terrains.
$A_{\text{terrain,habitat}}^{\text{effective impact}}$	Effective acres habitat impacted, that terrain, (acres)	$A_{\text{terrain}} \times u_{\text{mitigation}}$ for that terrain. Used in $C_{\text{credits,habitat}}$. Only when $M_{\text{terrain}} > 0$ for that terrain.
p_{wetland}	Wetland credit cost per acre, (\$/acre)	Single rate for all wetland acres.
$p_{\text{habitat}}(\text{terrain})$	Habitat credit cost per acre, (\$/acre)	Per terrain; explicit value for each habitat terrain.
$C_{\text{credits,wetlands}}$	Wetland credit cost (\$)	Use only when $M_{\text{terrain}} > 0$ for wetland.
$C_{\text{credits,habitat}}$	Habitat credit cost (\$)	Sum over habitat terrains of $p_{\text{habitat}}(\text{terrain}) \times A_{\text{terrain,habitat}}^{\text{effective impact}}$. Use only when $M_{\text{terrain}} > 0$ for habitat terrains.
$\xi_{\text{reconductoring}}$	Binary: not reconductoring, (0 or 1)	1 if not reconductoring, 0 if reconductoring. When 0, wetland and habitat credits set to zero.
$C_{\text{credits,total}}$	Total credit cost, wetland + habitat (\$)	After reconductoring rule.
T_{delay}	Delay years, (years)	
$T_{\text{construction}}$	Construction years, (years)	
t_{start}	Construction start year, (year index)	$t_{\text{start}} = T_{\text{delay}} + 1$ (1-based).
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Societal discount rate (§A.2).
$C_{\text{env.mit,nominal}}$	Total environmental mitigation cost, nominal (\$)	
$B_{\text{env.mit,annual}}$	Annual base for env. mitigation, (\$/year)	Over construction years (PV calculation).
C_{credits}	PV of credits (\$)	
$C_{\text{env.mit}}$	Total environmental mitigation cost (PV) (\$)	

Variables

Equations

1. Base area per terrain.

$$A_{\text{terrain}} = \frac{M_{\text{terrain}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

2. Effective acres (uplift for TCE).

$$A_{\text{terrain}}^{\text{effective}} = A_{\text{terrain}} \times u_{\text{mitigation}}.$$

3. Total base mitigation cost.

$$C_{\text{base,total}} = \sum_{\text{terrain}} c_{\text{base,peracre}}(\text{terrain}, CT) \times A_{\text{terrain}}^{\text{effective}}.$$

4. Wetland impact acres (wetland terrain only).

$$A_{\text{wetland}}^{\text{effective impact}} = A_{\text{wetland}} \times u_{\text{mitigation}}.$$

5. Habitat impact acres, total.

$$A_{\text{habitat}}^{\text{effective impact}} = \left(\sum_{\text{terrain} \in \text{habitat}} A_{\text{terrain}} \right) \times u_{\text{mitigation}}.$$

6. Wetland credit cost.

$$C_{\text{credits,wetlands}} = p_{\text{wetland}} \times A_{\text{wetland}}^{\text{effective impact}}.$$

7. Habitat credit cost.

$$C_{\text{credits,habitat}} = \sum_{\text{terrain} \in \text{habitat}} p_{\text{habitat}}(\text{terrain}) \times A_{\text{terrain,habitat}}^{\text{effective impact}}.$$

8. Total credit cost.

$$C_{\text{credits,total}} = \xi_{\text{reconductoring}} \times (C_{\text{credits,wetlands}} + C_{\text{credits,habitat}}).$$

9. Total nominal environmental mitigation cost.

$$C_{\text{env.mit,nominal}} = C_{\text{base,total}} + C_{\text{credits,total}}.$$

Regulatory perspective (AFUDC). Base mitigation and credits are AFUDC-eligible. The same capitalization logic as other capital costs applies (compound to COD; see §A.3). Total capitalized environmental mitigation equals capitalized base plus capitalized credits.

Societal perspective (present value). Base mitigation spread evenly over $T_{\text{construction}}$ starting at t_{start} . For $T_{\text{construction}} > 0$:

$$B_{\text{env.mit,annual}} = \frac{C_{\text{base,total}}}{T_{\text{construction}}}, \quad C_{\text{base}} = \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{B_{\text{env.mit,annual}}}{(1+r_{\text{WACC,real}})^t}.$$

If $T_{\text{construction}} = 0$, base is one-time at t_{start} : $C_{\text{base}} = C_{\text{base,total}}/(1+r_{\text{WACC,real}})^{t_{\text{start}}}$. Credits one-time at construction start (year t_{start}):

$$C_{\text{credits}} = \frac{C_{\text{credits,total}}}{(1+r_{\text{WACC,real}})^{t_{\text{start}}}}, \quad C_{\text{env.mit}} = C_{\text{base}} + C_{\text{credits}}.$$

Hard cost total

The three hard cost modules above produce the hard cost bucket. Hard costs are the one-time capital expenditures that enter rate base and, from the societal perspective, are discounted from construction spending.

$$C_{\text{hard}} = C_{\text{build}} + C_{\text{ROW,capital}} + C_{\text{env.mit}}.$$

Component PV equations: C_{build} (§B.1.1), $C_{\text{ROW,capital}}$ (§B.1.2), $C_{\text{env.mit}}$ (§B.1.3).

B.2 Soft Costs (C_{soft})

Soft costs are ongoing operational, energy-related, and time-value costs; none are AFUDC-eligible or in rate base. Three sub-aggregates: **operational** (O&M, insurance, ROW rent), **energy** (thermal loss, residual exceedance at $\gamma_{\text{congestion}}$), and **delay** (base delay, congestion delay, curtailment delay).

B.2.1 O&M

This subsection defines operations and maintenance cost: conductor, converter, structure, and vegetation management by terrain (terrain set as in §A.1), from commercial operation date over project lifetime.

Variable	Meaning / units	Notes
M_{total}	Total line length, (miles)	From physical details.
M_{terrain}	Miles in terrain type, (miles)	Terrain set as in §A.1. Only terrains with $M_{\text{terrain}} > 0$ used in sums.
T_{delay}	Delay years, (years)	Before construction.
$T_{\text{construction}}$	Construction years, (years)	
T_{lifetime}	Project lifetime, (years)	O&M accrues over this.
T_{COD}	Commercial operation date, (year index)	$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1$ (1-based; first year of O&M).
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount O&M (§A.2).
CT	Construction type, (index)	Overhead, Underground direct-buried, Underground tunnel, Subsea.
$c_{\text{conductor}}$	Conductor O&M, (\$/mile/year)	By category.
$c_{\text{converter}}$	Converter O&M, (\$/mile/year)	By category.
ξ_{DC}	Binary for DC projects, (0 or 1)	1 if DC, 0 if AC. Gates converter O&M (§B.1.1).
$n_{\text{structure(terrain)}}$	Structures per mile, (structures/mile)	By terrain. Overhead only.
$N_{\text{structure}}$	Total structures, (count)	Overhead: sum over terrains of $M_{\text{terrain}} \times n_{\text{structure(terrain)}}$. Non-overhead: see $c_{\text{structure,mi}}$.
$c_{\text{structure}}$	Cost per structure per year, (\$/structure/year)	Overhead only.
$c_{\text{structure,mi}}$	Structure O&M per mile per year, (\$/mile/year)	Non-overhead only; one value per construction type (often 0).
$c_{\text{veg(terrain)}}$	Vegetation management, (\$/mile/year)	By terrain. Overhead only.
$C_{\text{O\&M,annual}}$	Total annual O&M, (\$/year)	Sum of conductor + converter + structure + vegetation.
$C_{\text{O\&M,nominal}}$	O&M cost (nominal) (\$)	Undiscounted lifetime O&M.
$C_{\text{O\&M}}$	O&M cost (PV) (\$)	Present value of O&M from T_{COD} over T_{lifetime} .

Variables

Equations

1. COD (first year of O&M).

$$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1.$$

2. Conductor (annual).

$$C_{\text{conductor,annual}} = c_{\text{conductor}} \times M_{\text{total}}.$$

3. Converter (annual; DC only).

$$C_{\text{converter,annual}} = \xi_{\text{DC}} \times c_{\text{converter}} \times M_{\text{total}}.$$

4. Structure (annual).

$$C_{\text{structure,annual}} = \begin{cases} c_{\text{structure}} \times N_{\text{structure}} & \text{if overhead} \\ c_{\text{structure,mi}} \times M_{\text{total}} & \text{if non-overhead} \end{cases}, \quad N_{\text{structure}} = \sum_{\text{terrain}} M_{\text{terrain}} \times n_{\text{structure}}(\text{terrain}).$$

5. Vegetation management (annual).

$$C_{\text{veg,annual}} = \sum_{\text{terrain}} M_{\text{terrain}} \times c_{\text{veg}}(\text{terrain}).$$

6. Total annual O&M.

$$C_{\text{O\&M,annual}} = C_{\text{conductor,annual}} + C_{\text{converter,annual}} + C_{\text{structure,annual}} + C_{\text{veg,annual}}.$$

7. Nominal (undiscounted) lifetime O&M.

$$C_{\text{O\&M,nominal}} = C_{\text{O\&M,annual}} \times T_{\text{lifetime}}.$$

8. Real (present value) O&M.

$$C_{\text{O\&M}} = \sum_{t=T_{\text{COD}}}^{T_{\text{COD}}+T_{\text{lifetime}}-1} \frac{C_{\text{O\&M,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

B.2.2 Operational insurance

This subsection defines operational insurance cost: annual premiums on insurable asset value (build-cost components) from COD to end of life, discounted to present value at real WACC.

Variable	Meaning / units	Notes
$C_{\text{adj,cont,conductor}}$	Conductor build cost (\$)	Terrain-adjusted, contingency applied. From Build costs (§B.1.1).
$C_{\text{adj,cont,structure}}$	Structure build cost (\$)	Terrain-adjusted, contingency applied. From Build costs (§B.1.1).
$C_{\text{adj,cont,converter}}$	Converter build cost (\$)	Terrain-adjusted, contingency applied. From Build costs (§B.1.1).
ξ_{DC}	Binary for DC projects, (0 or 1)	1 if DC, 0 if AC. Gates converter in insurable value (§B.1.1).
$\xi_{\text{ins,conductor}}$	Include conductors in insurable value, (0 or 1)	1 if insured, 0 if not.
$\xi_{\text{ins,structure}}$	Include structures in insurable value, (0 or 1)	1 if insured, 0 if not.
$\xi_{\text{ins,converter}}$	Include converters in insurable value, (0 or 1)	1 if insured, 0 if not.
$V_{\text{insurable}}$	Insurable asset value (\$)	Sum of included build-cost components (with ξ_{DC} for converter).
CT	Construction type, (index)	Overhead, Underground direct-buried, Underground tunnel, Subsea.
$r_{\text{premium}}(CT)$	Premium rate, (decimal)	Annual premium as fraction of insurable value. Can depend on CT .
$r_{\text{premium,default}}$	Default premium rate, (decimal)	Used when no type-specific rate is defined for CT .
$C_{\text{insurance,annual}}$	Annual operational insurance premium, (\$/year)	
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date, (year index)	First year premium is paid. $T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1$.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount premiums (§A.2).
$C_{\text{insurance,nominal}}$	Operational insurance cost (nominal) (\$)	Undiscounted sum of premiums over project lifetime.
$C_{\text{insurance}}$	Operational insurance cost (PV) (\$)	Present value of premium stream from T_{COD} over T_{lifetime} .

Variables

Equations

1. Insurable asset value.

$$V_{\text{insurable}} = \xi_{\text{ins,conductor}} \cdot C_{\text{adj,cont,conductor}} + \xi_{\text{ins,structure}} \cdot C_{\text{adj,cont,structure}} + \xi_{\text{ins,converter}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}.$$

2. Premium rate.

$$r_{\text{premium}}(CT) = \begin{cases} r_{\text{premium},CT} & \text{if a rate is specified for construction type } CT \\ r_{\text{premium,default}} & \text{otherwise} \end{cases}$$

3. Annual premium.

$$C_{\text{insurance,annual}} = V_{\text{insurable}} \times r_{\text{premium}}(CT).$$

4. Nominal (undiscounted) lifetime cost.

$$C_{\text{insurance,nominal}} = C_{\text{insurance,annual}} \times T_{\text{lifetime}}.$$

5. Real (present value) cost.

$$C_{\text{insurance}} = \sum_{t=T_{\text{COD}}}^{T_{\text{COD}}+T_{\text{lifetime}}-1} \frac{C_{\text{insurance,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

B.2.3 Operational ROW rent

This subsection defines operational ROW rent: annual payment for use of the right-of-way in zones where the line is built. Same zones, areas, and per-acre rent as in Capital ROW (§B.1.2); agreement type determines whether rent applies and over which years.

Variable	Meaning / units	Notes
zone	Zone index, (index)	Same zones as Capital ROW; use only zones where $M_{\text{zone}} > 0$.
M_{zone}	Miles in zone, (miles)	
W_{ROW}	ROW width, (ft)	Same as Capital ROW (project category).
A_{zone}	Area of ROW of zone, (acres)	$A_{\text{zone}} = (M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}})/43\,560 \text{ (ft}^2\text{/acre)}$.
$p_{\text{rent,zone}}$	Rent per acre per year, zone, (\$/acre/year)	
$C_{\text{rent,annual}}$	Total annual ROW rent, (\$/year)	Sum over zones.
ξ_{rent}	Binary: rent applies, (0 or 1)	1 for Existing Lease/License or Federal Hybrid; 0 for New Permanent Easement or Simple Fee.
T_{delay}	Delay years, (years)	
$T_{\text{construction}}$	Construction years, (years)	
T_{lifetime}	Project lifetime, (years)	Years of operation.
$t_{\text{rent,start}}$	First year rent is paid, (year index)	Agreement-dependent (see table).
T_{rent}	Number of years rent is paid, (years)	Agreement-dependent (see table).
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount rent (§A.2).
$C_{\text{rent,total,nominal}}$	ROW rent cost (nominal) (\$)	Undiscounted sum over all years rent is paid.
$C_{\text{rent,total}}$	ROW rent cost (PV) (\$)	Present value of rent stream.

Agreement type		Rent?	$t_{\text{rent,start}}$	T_{rent}
New	Permanent	No ($\xi_{\text{rent}} = 0$)	—	—
Easement				
Simple Fee		No ($\xi_{\text{rent}} = 0$)	—	—
Existing		Yes ($\xi_{\text{rent}} = 1$)	1	$T_{\text{delay}} + T_{\text{construction}} + T_{\text{lifetime}}$
Lease/License				
Federal Hybrid		Yes ($\xi_{\text{rent}} = 1$)	$T_{\text{delay}} + 1$	$T_{\text{construction}} + T_{\text{lifetime}}$

Variables

Equations

1. Area of ROW per zone.

$$A_{\text{zone}} = \frac{M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

2. Annual ROW rent.

$$C_{\text{rent,annual}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{rent,zone}}.$$

3. Nominal (undiscounted) ROW rent cost.

$$C_{\text{rent,total,nominal}} = \xi_{\text{rent}} \times C_{\text{rent,annual}} \times T_{\text{rent}}.$$

4. Real (present value) ROW rent cost.

$$C_{\text{rent,total}} = \xi_{\text{rent}} \times \sum_{t=t_{\text{rent,start}}}^{t_{\text{rent,start}}+T_{\text{rent}}-1} \frac{C_{\text{rent,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

B.2.4 Energy losses

This subsection defines energy losses: the physical quantity of energy lost on the transmission path (line and, for DC, converters), used for energy loss costs (§B.2.5) and emissions costs (§B.4).

Variable	Meaning / units	Notes
C_{new}	New line capacity, (MW)	Nameplate. Reconductoring and greenfield.
V	Line voltage, (kV)	Reconductoring: existing (old) config; greenfield: new config.
L	Line length, (miles)	Total line length.
u	Line utilization, (dimensionless)	$0 \leq u \leq 1$. Average fraction of capacity used.
n_{phases}	Number of phases, (count)	3 for AC; 2 for DC (poles).
n_{circuits}	Number of circuits / poles, (count)	Parallel circuits (AC) or poles (DC).
$n_{\text{conductorsperphase}}$	Conductors per phase (or per pole), (count)	Conductors in parallel per phase/pole.
R_{mi}	Resistance per mile, (Ω/mile)	AC: 75°C; DC: 20°C. Depends on conductor and circuit.
I	Phase current, (A)	RMS phase current at full capacity and nominal voltage.
ϕ_{AC}	AC power factor, (dimensionless)	e.g. 0.95. AC only.
f_{load}	Full-load adjustment factor, (dimensionless)	Function of u ; converts full-load loss to average over utilization.
H	Hours per year, (h/year)	e.g. 8760.
T_{lifetime}	Project lifetime, (years)	
ξ_{DC}	Binary for DC, (0 or 1)	1 if DC, 0 if AC.
N_{conv}	Number of converter stations, (count)	0 for AC; typically 2 for DC.
λ_{conv}	Converter loss fraction, (dimensionless)	Fraction of through-power lost per station (e.g. LCC, VSC).
$P_{\text{lineloss,MW}}$	Line loss (power), (MW)	Total resistive line loss (average power).
$E_{\text{lineloss,MW,annual}}$	Line energy losses (annual), (MWh/year)	
$E_{\text{lineloss,lifetime}}$	Line energy losses (lifetime), (MWh)	
$P_{\text{converterloss,MW}}$	Converter loss (power), (MW)	DC only.
$E_{\text{converterloss,MW,annual}}$	Converter energy losses (annual), (MWh/year)	
E_{loss}	Total energy losses (annual), (MWh/year)	Line + converter.
$E_{\text{totalloss,lifetime}}$	Total energy losses (lifetime), (MWh)	

Variables

Equations

1. Phase current (from capacity and voltage).

$$I = \frac{C_{\text{new}} \times 1000 \text{ (A}\cdot\text{kV/MW)}}{\phi_{\text{AC}} \times V \times \sqrt{n_{\text{phases}}} \times n_{\text{circuits}}} \quad (\text{AC}), \quad I = \frac{C_{\text{new}} \times 1000 \text{ (A}\cdot\text{kV/MW)}}{V \times \sqrt{n_{\text{phases}}} \times n_{\text{circuits}}} \quad (\text{DC}).$$

2. Full-load adjustment (utilization).

$$f_{\text{load}} = \frac{u + u^2}{2}.$$

3. Line loss (power). Total resistive (I^2R) line loss, average power in MW:

$$P_{\text{lineloss,MW}} = \left(\frac{I}{n_{\text{conductorsperphase}}} \right)^2 \times (n_{\text{conductorsperphase}} \times n_{\text{phases}} \times n_{\text{circuits}}) \times R_{\text{mi}} \times L \times \frac{f_{\text{load}}}{10^6}.$$

4. Line energy losses (annual and lifetime).

$$E_{\text{lineloss,MW,annual}} = P_{\text{lineloss,MW}} \times H, \quad E_{\text{lineloss,lifetime}} = E_{\text{lineloss,MW,annual}} \times T_{\text{lifetime}}.$$

5. Converter loss (DC only).

$$P_{\text{converterloss,MW}} = \xi_{\text{DC}} \times N_{\text{conv}} \times \lambda_{\text{conv}} \times u \times C_{\text{new}}, \quad E_{\text{converterloss,MW,annual}} = P_{\text{converterloss,MW}} \times H.$$

6. Total energy losses.

$$E_{\text{loss}} = E_{\text{lineloss,MW,annual}} + E_{\text{converterloss,MW,annual}}, \quad E_{\text{totalloss,lifetime}} = E_{\text{loss}} \times T_{\text{lifetime}}.$$

B.2.5 Thermal loss costs

This subsection defines thermal loss cost: the cost of the energy lost on the transmission path (line and, for DC, converters), valuing energy losses from §B.2.4 at the value of load—the demand-side marginal value of the energy that would have been delivered.

Variable	Meaning / units	Notes
E_{loss}	Total energy losses (annual), (MWh/year)	From §B.2.4 (line + converter).
$E_{\text{lineloss,MW,annual}}$	Line energy losses (annual), (MWh/year)	Conductor only.
$E_{\text{converterloss,MW,annual}}$	Converter energy losses (annual), (MWh/year)	DC only.
v_{load}	Value of load, (\$/MWh)	Marginal value of energy to end-use load. The value of electricity derives from the services load performs with it, making demand-side valuation the correct welfare measure.
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date, (year)	First year of operation.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount this cost (§A.2).
$C_{\text{lineloss,annual}}$	Line loss cost (annual), (\$/year)	Cost of line losses only.
$C_{\text{converterloss,annual}}$	Converter loss cost (annual), (\$/year)	Cost of converter losses only.
$C_{\text{loss,annual}}$	Total thermal loss cost (annual), (\$/year)	Line + converter.
$C_{\text{loss,nominal}}$	Total thermal loss cost (nominal) (\$)	Undiscounted sum over lifetime.
C_{loss}	Total thermal loss cost (PV) (\$)	PV of annual stream from T_{COD} over T_{lifetime} .

Variables

Equations

1. Component costs (line and converter).

$$C_{\text{lineloss,annual}} = E_{\text{lineloss,MW,annual}} \times v_{\text{load}}, \quad C_{\text{converterloss,annual}} = E_{\text{converterloss,MW,annual}} \times v_{\text{load}}.$$

2. Total annual thermal loss cost.

$$C_{\text{loss,annual}} = E_{\text{loss}} \times v_{\text{load}} = C_{\text{lineloss,annual}} + C_{\text{converterloss,annual}}.$$

3. Nominal (undiscounted) lifetime cost.

$$C_{\text{loss,nominal}} = C_{\text{loss,annual}} \times T_{\text{lifetime}}.$$

4. Real (present value) cost.

$$C_{\text{loss}} = \sum_{t=T_{\text{COD}}}^{T_{\text{COD}}+T_{\text{lifetime}}-1} \frac{C_{\text{loss,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

Residual exceedance cost

If the project does not fully relieve the constraint (i.e. $X_{\text{congestion}} > \Delta C_{\text{effective}}$ or $MW_{\text{curtailment}} > \Delta C_{\text{effective}}$), the remaining congestion and/or curtailment energy is valued at $\gamma_{\text{congestion}}$:

$$E_{\text{residual}} = H_{\text{congestion}} \times \max(0, X_{\text{congestion}} - \Delta C_{\text{effective}}) + H_{\text{curtailment}} \times \max(0, MW_{\text{curtailment}} - \Delta C_{\text{effective}}),$$

$$C_{\text{residual,annual}} = E_{\text{residual}} \times \gamma_{\text{congestion}}.$$

$C_{\text{residual,nominal}}$ and C_{residual} (PV) follow the same discounting convention as thermal loss costs (level annuity from T_{COD} over T_{lifetime} , discounted at $r_{\text{WACC,real}}$).

Energy loss cost aggregate

$$C_{\text{energy}} = C_{\text{loss}} + C_{\text{residual}}.$$

B.2.6 Base delay costs

This subsection defines base delay costs: the annual costs incurred during the delay (permitting and pre-construction) period (legal, administrative, labor, materials and equipment, regulatory, public relations, project management, miscellaneous). They are expressed as a constant annual cost for each delay year. Total nominal cost is that annual cost times the number of delay years. Present value is a level annuity over the delay period, discounted at real WACC.

Variable	Meaning / units	Notes
$k \in \mathcal{K}$	Delay cost category index, (index)	$\mathcal{K}$ is the set of delay cost categories: legal, admin, labor, material and equipment, regulatory, public relations, project management, miscellaneous.
$C_{\text{delay},k}$	Annual delay cost for category k , (\$/year)	$C_{\text{delay,annual}} = \sum_{k \in \mathcal{K}} C_{\text{delay},k}$.
$C_{\text{delay,annual}}$	Total annual delay cost, (\$/year)	Sum over categories.
T_{delay}	Delay years, (years)	From project technical details; permitting/pre-construction period.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount the delay cost stream (§A.2).
$C_{\text{delay,nominal}}$	Total base delay cost (nominal) (\$)	$C_{\text{delay,nominal}} = C_{\text{delay,annual}} \times T_{\text{delay}}$.
$C_{\text{base,delay}}$	Total base delay cost (PV) (\$)	PV of level annuity over T_{delay} years at $r_{\text{WACC,real}}$.

Variables

Equations

1. Total annual delay cost.

$$C_{\text{delay,annual}} = \sum_{k \in \mathcal{K}} C_{\text{delay},k}.$$

2. Nominal total base delay cost.

$$C_{\text{delay,nominal}} = C_{\text{delay,annual}} \times T_{\text{delay}}.$$

3. Real (present value) base delay cost. Level annuity over years $t = 1, \dots, T_{\text{delay}}$, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{base,delay}} = C_{\text{delay,annual}} \times \frac{1 - (1 + r_{\text{WACC,real}})^{-T_{\text{delay}}}}{r_{\text{WACC,real}}}.$$

Link to other delay cost categories. Congestion delay costs (§B.2.7) and curtailment delay costs (§B.2.8) are opportunity costs of foregone congestion/curtailment relief during delay and construction; see those sections. All three delay cost categories apply only during the pre-COD period (delay and construction); the benefit streams in §C apply only post-COD. A given year falls in exactly one of these periods, so there is no double-count between delay costs and benefits.

B.2.7 Congestion delay costs

This subsection defines congestion delay costs: the opportunity cost of congestion that is not relieved while the project is in the delay and construction period. In each of those years, the same congestion relief (MWh/year) that the project would provide once in service is valued at the same price (\$/MWh); that annual value is the cost of delay. The annual congestion delay cost equals the annual congestion reduction benefit. The cost stream runs over $T_{\text{delay}} + T_{\text{construction}}$ years; present value is a level annuity over that period, discounted at real WACC. Conceptually: this cost answers “What congestion benefit do we give up each year we are delayed?” Capacity relief and allocation (curtailment first, then congestion; overlap handled) are the same as in §C. The same annual \$ amount is used as the congestion benefit, so there is no double-count with post-COD benefits.

Variable	Meaning / units	Notes
$B_{\text{congestion,annual}}$	Annual congestion reduction benefit (\$/year)	Same formula as in §C: allocated congestion relief MWh/year $\times \gamma_{\text{congestion}}$.
$C_{\text{cong,delay,annual}}$	Congestion delay cost (annual), (\$/year)	$C_{\text{cong,delay,annual}} = B_{\text{congestion,annual}}$.
T_{delay}	Delay years	From project technical details.
$T_{\text{construction}}$	Construction years	From project technical details.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount the delay cost stream (§A.2).
$C_{\text{cong,delay,nominal}}$	Congestion delay cost (nominal) (\$)	Total over delay + construction; undiscounted.
$C_{\text{cong,delay}}$	Congestion delay cost (PV) (\$)	PV of level annuity over $T_{\text{delay}} + T_{\text{construction}}$ years at $r_{\text{WACC,real}}$, starting at year 1.

Variables

Equations

1. Annual congestion delay cost. The annual opportunity cost equals the annual congestion reduction benefit (allocated congestion relief $\times \gamma_{\text{congestion}}$; see §C):

$$C_{\text{cong,delay,annual}} = B_{\text{congestion,annual}}.$$

2. Nominal total congestion delay cost.

$$C_{\text{cong,delay,nominal}} = C_{\text{cong,delay,annual}} \times (T_{\text{delay}} + T_{\text{construction}}).$$

3. Real (present value) congestion delay cost. Level annuity over years $t = 1, \dots, T_{\text{delay}} + T_{\text{construction}}$, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{cong,delay}} = C_{\text{cong,delay,annual}} \times \frac{1 - (1 + r_{\text{WACC,real}})^{-(T_{\text{delay}} + T_{\text{construction}})}}{r_{\text{WACC,real}}}.$$

Link to Congestion and curtailment reduction (§C). The same effective capacity relief, curtailment-then-congestion allocation, and overlap logic define $B_{\text{congestion,annual}}$. This subsection uses that annual value as the cost per year of delay/construction, so benefits (post-COD) and congestion delay cost (during delay/construction) are consistent and not double-counted.

Link to Curtailment delay costs (§B.2.8). Curtailment delay costs and this subsection use the same period ($T_{\text{delay}} + T_{\text{construction}}$) and discounting ($r_{\text{WACC,real}}$). Allocation is curtailment first, then congestion, so the two delay costs are consistent and not double-counted.

Link to Base delay costs (§B.2.6). Base delay costs are out-of-pocket delay costs (legal, admin, etc.). Congestion delay costs are the opportunity cost of foregone congestion relief during the same period. Both use T_{delay} ; this subsection uses $T_{\text{delay}} + T_{\text{construction}}$ as the cost duration and $r_{\text{WACC,real}}$ for PV.

B.2.8 Curtailment delay costs

This subsection defines curtailment delay costs: the opportunity cost of curtailment that is not relieved while the project is in the delay and construction period. In each of those years, the same curtailment relief (MWh/year) that the project would provide once in service is valued at the same price (\$/MWh); that annual value is the cost of delay. The annual curtailment delay cost equals the annual curtailment reduction benefit. The cost stream runs over $T_{\text{delay}} + T_{\text{construction}}$ years; present value is a level annuity over that period, discounted at real WACC. Conceptually: this cost answers “What curtailment benefit do we give up each year we are delayed?” Capacity relief and allocation (curtailment first, then congestion) are the same as in §C. The same annual \$ amount is used as the curtailment benefit, so there is no double-count with post-COD benefits.

Variable	Meaning / units	Notes
$B_{\text{curtailment,annual}}$	Annual curtailment reduction benefit (\$/year)	Same formula as in §C: allocated curtailment relief MWh/year $\times \gamma_{\text{curtailment}}$.
$C_{\text{curt,delay,annual}}$	Curtailment delay cost (annual), (\$/year)	$C_{\text{curt,delay,annual}} = B_{\text{curtailment,annual}}$.
T_{delay}	Delay years	From project technical details.
$T_{\text{construction}}$	Construction years	From project technical details.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount the delay cost stream (§A.2).
$C_{\text{curt,delay,nominal}}$	Curtailment delay cost (nominal) (\$)	Total over delay + construction; undiscounted.
$C_{\text{curt,delay}}$	Curtailment delay cost (PV) (\$)	PV of level annuity over $T_{\text{delay}} + T_{\text{construction}}$ years at $r_{\text{WACC,real}}$, starting at year 1.

Variables

Equations

1. Annual curtailment delay cost. The annual opportunity cost equals the annual curtailment reduction benefit (allocated curtailment relief $\times \gamma_{\text{curtailment}}$; see §C):

$$C_{\text{curt,delay,annual}} = B_{\text{curtailment,annual}}.$$

2. Nominal total curtailment delay cost.

$$C_{\text{curt,delay,nominal}} = C_{\text{curt,delay,annual}} \times (T_{\text{delay}} + T_{\text{construction}}).$$

3. Real (present value) curtailment delay cost. Level annuity over years $t = 1, \dots, T_{\text{delay}} + T_{\text{construction}}$, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{curt,delay}} = C_{\text{curt,delay,annual}} \times \frac{1 - (1 + r_{\text{WACC,real}})^{-(T_{\text{delay}} + T_{\text{construction}})}}{r_{\text{WACC,real}}}.$$

Link to Congestion and curtailment reduction (§C). The same effective capacity relief and curtailment-then-congestion allocation define $B_{\text{curtailment,annual}}$. This subsection uses that annual value as the cost per year of delay/construction, so benefits (post-COD) and curtailment delay cost (during delay/construction) are consistent and not double-counted.

Link to Congestion delay costs (§B.2.7). Congestion delay costs and this subsection use the same period ($T_{\text{delay}} + T_{\text{construction}}$) and discounting ($r_{\text{WACC,real}}$). Allocation is curtailment first, then congestion, so the two delay costs are consistent and not double-counted.

Link to Base delay costs (§B.2.6). Base delay costs are out-of-pocket delay costs (legal, admin, etc.). Curtailment delay costs are the opportunity cost of foregone curtailment relief during the same delay and construction period.

Soft cost total

The nine soft cost modules above produce the soft cost bucket. $C_{\text{emissions}}$ is a separate bucket (§B.4); see also §D. Soft costs are ongoing operational, energy-related, and time-value costs; none are AFUDC-eligible or in rate base.

$$\begin{aligned} C_{\text{operational}} &= C_{\text{O\&M}} + C_{\text{insurance}} + C_{\text{rent,total}}, \\ C_{\text{energy}} &= C_{\text{loss}} + C_{\text{residual}}, \\ C_{\text{delay}} &= C_{\text{base,delay}} + C_{\text{cong,delay}} + C_{\text{curt,delay}}, \\ C_{\text{soft}} &= C_{\text{operational}} + C_{\text{energy}} + C_{\text{delay}}. \end{aligned}$$

Component PV equations: $C_{\text{O\&M}}$ (§B.2.1), $C_{\text{insurance}}$ (§B.2.2), $C_{\text{rent,total}}$ (§B.2.3), C_{loss} (§B.2.5), C_{residual} (§B.2.5), $C_{\text{base,delay}}$ (§B.2.6), $C_{\text{cong,delay}}$ (§B.2.7), $C_{\text{curt,delay}}$ (§B.2.8).

B.3 Risk Costs (C_{risk})

This subsection defines risk costs: expected cost of wildfires and expected cost of outages. None of these costs are AFUDC-eligible or included in rate base. These two sub-items produce the risk cost bucket C_{risk} (§B.3.2).

B.3.1 Expected cost of wildfires

This subsection defines expected cost of wildfires: the expected annual loss (EAL) from wildfire events over the project's operating life. It uses ignition rates by terrain and construction type, multiplies by severity (\$ per event), and can apply risk growth over the lifetime. Present value is a growing annuity from COD, typically discounted at the social discount rate.

Variable	Meaning / units	Notes
terrain	Terrain index	Terrain set as in §A.1.
L_{terrain}	Line length in terrain, (miles)	From physical details.
$f_{\text{base,terrain}}$	Base ignition rate for terrain, (events/(mi·yr))	Overhead baseline.
k_t	Construction-type multiplier, (dimensionless)	Overhead 1; underground < 1; subsea 0.
$f_{\text{terrain},t}$	Effective ignition rate for terrain and type t , (events/(mi·yr))	$f_{\text{terrain},t} = f_{\text{base,terrain}} \times k_t$.
λ_{terrain}	Annual event rate for terrain, (events/year)	$\lambda_{\text{terrain}} = L_{\text{terrain}} \times f_{\text{terrain},t}$.
λ	Total annual event rate, (events/year)	$\lambda = \sum_{\text{terrain}} \lambda_{\text{terrain}}$.
S	Severity (loss per event) (\$)	Expected loss per wildfire event.
EAL	Expected annual loss, (\$/year)	$EAL = \lambda \times S$.
g_{wf}	Wildfire risk growth rate, (decimal)	Annual increase in expected loss (e.g. escalation of risk).
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date (year index), (years)	First year of operation; cost stream starts here.
T_{delay}	Delay years	Used in PV discounting.
$T_{\text{construction}}$	Construction years	Used in PV discounting.
r_{social}	Social discount rate, (decimal)	Used to discount this externality cost (§A.2).
$C_{\text{wf,nominal}}$	Expected wildfire cost (nominal) (\$)	Sum of growing annual EAL over lifetime; undiscounted.
C_{wf}	Expected wildfire cost (PV) (\$)	PV of growing EAL stream from T_{COD} over T_{lifetime} ; discounted for delay and construction.

Variables

Equations

1. Effective ignition rate (terrain and construction type).

$$f_{\text{terrain},t} = f_{\text{base,terrain}} \times k_t.$$

2. Segment and total annual event rate.

$$\lambda_{\text{terrain}} = L_{\text{terrain}} \times f_{\text{terrain},t}, \quad \lambda = \sum_{\text{terrain}} \lambda_{\text{terrain}}.$$

3. Expected annual loss.

$$EAL = \lambda \times S.$$

4. Nominal (undiscounted) total cost. Growing annual loss at rate g_{wf} over T_{lifetime} years:

$$C_{\text{wf,nominal}} = EAL \times \frac{(1 + g_{\text{wf}})^{T_{\text{lifetime}}} - 1}{g_{\text{wf}}}.$$

5. Real (present value) cost. Two reference dates: Year 0 = project analysis start (permitting, pre-construction); T_{COD} = commercial operation date = first year the cost stream begins ($T_{\text{delay}} + T_{\text{construction}}$ years after Year 0). Present value is expressed in Year-0 dollars.

Step 1: The numerator is the present value of the growing annuity of annual losses (EAL growing at g_{wf}) over T_{lifetime} years, with “present” taken at T_{COD} . So it is the lump sum at T_{COD} equivalent to the stream from T_{COD} to end of life. Value at T_{COD} :

$$\text{PV at } T_{\text{COD}} = EAL \times \frac{1 - \left(\frac{1+g_{\text{wf}}}{1+r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{wf}}}.$$

Step 2: The denominator moves that lump sum from T_{COD} back to Year 0: the amount at T_{COD} is discounted by $(1 + r_{\text{social}})^{T_{\text{delay}}+T_{\text{construction}}}$.

The real (present value) cost is therefore:

$$C_{\text{wf}} = \frac{EAL \times \frac{1 - \left(\frac{1+g_{\text{wf}}}{1+r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{wf}}}}{(1 + r_{\text{social}})^{T_{\text{delay}}+T_{\text{construction}}}}.$$

B.3.2 Expected cost of outages

This subsection defines expected cost of outages: the expected annual cost (EAC) from transmission outages over the project's operating life. It uses outage rates by terrain and construction type, effective duration (hours per event) by terrain and type, capacity at risk (fraction of line capacity lost per event), and value of lost load (VoLL) piecewise by duration. Present value is a growing annuity from COD, typically discounted at the social discount rate. The cost stream starts at COD (after delay and construction).

Variable	Meaning / units	Notes
terrain	Terrain index	e.g. forested, wetland, urban.
CT	Construction type index	e.g. overhead, underground, subsea.
L_{terrain}	Line length in terrain, (miles)	From physical details.
$r_{\text{terrain},CT}$	Outage rate for terrain and construction type CT , (events/(mi·yr))	Direct outages per mile per year.
λ_{terrain}	Annual outage rate for terrain, (events/year)	$\lambda_{\text{terrain}} = L_{\text{terrain}} \times r_{\text{terrain},CT}$.
λ	Total annual outage rate, (events/year)	$\lambda = \sum_{\text{terrain}} \lambda_{\text{terrain}}$.
$H_{\text{base},\text{terrain}}$	Base outage duration for terrain, (h/event)	Hours per outage event (by terrain).
k_{CT}	Duration multiplier for construction type CT , (dimensionless)	Overhead 1; underground, subsea can be > 1 .
$H_{\text{eff},\text{terrain}}$	Effective outage duration for terrain (and construction type CT), (h/event)	$H_{\text{eff},\text{terrain}} = H_{\text{base},\text{terrain}} \times k_{CT}$.
ϕ	Capacity-at-risk factor, (dimensionless)	Fraction of line capacity lost per event; e.g. 1 = radial.
C	Line capacity, (MW)	Project capacity.
MW_{lost}	MW lost per event, (MW)	$MW_{\text{lost}} = \phi \times C$.
U_{terrain}	Unserved energy per event (terrain), (MWh/event)	$U_{\text{terrain}} = H_{\text{eff},\text{terrain}} \times MW_{\text{lost}}$.
v_1, v_2, v_3 (v_{tier})	VoLL by duration tier, (\$/MWh)	Tier 1: 0–4 h, tier 2: 4–24 h, tier 3: 24+ h.
$C_{\text{event},\text{terrain}}$	Cost per outage event (terrain), \$/event	Piecewise VoLL over U_{terrain} (unserved MWh valued at v_{tier} by duration tier).
EAC	Expected annual cost, (\$/year)	$EAC = \sum_{\text{terrain}} \lambda_{\text{terrain}} \times C_{\text{event},\text{terrain}}$.
g_{outages}	Outage risk growth rate, (decimal)	Annual increase in expected cost (optional).
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date (year index), (years)	First year of operation; cost stream starts here.
T_{delay}	Delay years	Used in PV discounting.
$T_{\text{construction}}$	Construction years	Used in PV discounting.
r_{social}	Social discount rate, (decimal)	Used to discount this externality cost (§A.2).
$C_{\text{outage,nominal}}$	Expected outage cost (nominal) (\$)	Sum of growing annual EAC over lifetime; undiscounted.
C_{outage}	Expected outage cost (PV) (\$)	PV of growing EAC stream from T_{COD} over T_{lifetime} ; discounted for delay and construction.

Variables

Equations

1. Outage rate and effective duration (by terrain and construction type).

$$\lambda_{\text{terrain}} = L_{\text{terrain}} \times r_{\text{terrain},CT}, \quad H_{\text{eff,terrain}} = H_{\text{base,terrain}} \times k_{CT}.$$

2. Total annual outage rate.

$$\lambda = \sum_{\text{terrain}} \lambda_{\text{terrain}}.$$

3. MW lost and unserved energy per event (terrain).

$$MW_{\text{lost}} = \phi \times C, \quad U_{\text{terrain}} = H_{\text{eff,terrain}} \times MW_{\text{lost}}.$$

4. Cost per event (piecewise VoLL). Duration is split into tiers with VoLL (\$/MWh) per tier:

Duration tier	VoLL
0–4 h	v_1
4–24 h	v_2
24+ h	v_3

Cost per event is the sum over tiers of (hours in that tier) $\times MW_{\text{lost}} \times v_{\text{tier}}$:

$$C_{\text{event,terrain}} = \sum_{\text{tiers}} (\text{hours in tier}) \times MW_{\text{lost}} \times v_{\text{tier}}.$$

5. Expected annual cost.

$$EAC = \sum_{\text{terrain}} \lambda_{\text{terrain}} \times C_{\text{event,terrain}}.$$

6. Nominal (undiscounted) total cost. Growing annual cost at rate g_{outages} over T_{lifetime} years:

$$C_{\text{outage,nominal}} = EAC \times \frac{(1 + g_{\text{outages}})^{T_{\text{lifetime}}} - 1}{g_{\text{outages}}}.$$

7. Real (present value) cost. Two reference dates: Year 0 = project analysis start (permitting, pre-construction); T_{COD} = commercial operation date = first year of operation ($T_{\text{delay}} + T_{\text{construction}}$ years after Year 0). Present value is expressed in Year-0 dollars.

Step 1: The numerator is the present value of the growing annuity of annual outage costs (EAC growing at g_{outages}) over T_{lifetime} years, as of T_{COD} —the lump sum at T_{COD} equivalent to the stream from T_{COD} to end of life. Value at T_{COD} :

$$\text{PV at } T_{\text{COD}} = EAC \times \frac{1 - \left(\frac{1+g_{\text{outages}}}{1+r_{\text{social}}}\right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{outages}}}.$$

Step 2: The denominator discounts that lump sum from T_{COD} back to Year 0 by $(1 + r_{\text{social}})^{T_{\text{delay}}+T_{\text{construction}}}$.

The real (present value) cost is therefore:

$$C_{\text{outage}} = \frac{EAC \times \frac{1 - \left(\frac{1+g_{\text{outages}}}{1+r_{\text{social}}}\right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{outages}}}}{(1 + r_{\text{social}})^{T_{\text{delay}}+T_{\text{construction}}}}.$$

Note on VoLL. Value of lost load (VoLL) is the economic value placed on unserved energy (\$/MWh). Tiered VoLL reflects higher marginal value for longer outages (e.g. short 0–4 h, medium 4–24 h, long 24+ h). Capacity at risk ϕ (e.g. 1 for radial, < 1 for meshed/redundant) is the fraction of line capacity assumed lost per outage event.

Risk cost total

The two risk cost modules above produce the risk cost bucket. Risk costs are probabilistic and event-driven; they are not AFUDC-eligible and do not enter rate base.

$$C_{\text{risk}} = C_{\text{wf}} + C_{\text{outage}}.$$

Component PV equations: C_{wf} (§B.3.1), C_{outage} (§B.3.2).

B.4 Emissions Costs ($C_{\text{emissions}}$)

This subsection defines the two cost components of the emissions bucket: **loss-compensation emissions** ($C_{\text{comp,emissions}}$) and **facilitated emissions** ($C_{\text{fac,emissions}}$). Both use the same rate-based mix-evolution engine, emission intensities $I_{j,k}$, societal costs c_k , and social discount rate r_{social} . The emissions cost bucket is $C_{\text{emissions}}^P = C_{\text{comp,emissions}}^P + C_{\text{fac,emissions}}^P$. The displacement avoided cost $C_{\text{displ,emissions}}^P$ (§C.3) is computed and reported but does not enter NB or BCR; the clean-generation incentive appears through $\Delta C_{\text{emissions}}^{P:B}$ in pairwise comparisons. None of these costs are AFUDC-eligible or included in rate base.

B.4.1 Loss-compensation emissions

This subsection defines $C_{\text{comp,emissions}}$: the societal cost of emissions (CO_2 , SO_x , NO_x) from extra generation used to compensate for transmission energy losses. Energy losses (E_{loss}) are defined in §B.2.4; this subsection only adds externality valuation through the line's time-varying fuel mix and pollutant prices. Shares $p_j(\tau)$ here are the **project-path** shares $p_j^{\text{proj}}(\tau)$ from **energy_source_mix**; the counterfactual path $p_j^{\text{no}}(\tau)$ used by the facilitated-emissions subsection (§B.4.2) does not enter loss-compensation calculations.

Variable	Meaning / units	Notes
E_{loss}	Total energy losses (annual), (MWh/year)	From §B.2.4.
α	Compensation share, (dimensionless [0,1])	Fraction of E_{loss} compensated by additional generation.
TEC	Total energy compensated (annual), (MWh/year)	$TEC = \alpha \times E_{\text{loss}}$.
j	Energy source index, (index)	e.g. coal, gas, solar, wind, hydro, nuclear.
$p_j^{\text{proj}}(\tau)$	Project-path share of source j in year τ , (dimensionless)	From energy_source_mix initial mix and growth/decay; $\sum_j p_j^{\text{proj}}(\tau) = 1$. See §B.4.2 for the counterfactual path.
$I_{j,k}$	Emission intensity, (kg/MWh)	Emissions of pollutant k per MWh from source j .
k	Pollutant index, (index)	CO_2 , SO_x , NO_x .
$E_{k,\tau}$	Emissions of pollutant k in year τ , (kg)	From compensated energy and mix.
c_k	Societal cost per kg of pollutant k , (\$/kg)	Externality value.
$C_{k,\tau}$	Cost of pollutant k in year τ (\$)	$E_{k,\tau} \times c_k$.
C_τ	Total emissions cost in year τ , (\$/year)	Sum over pollutants.
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date, (year)	First year of operation.
r_{social}	Social discount rate, (decimal)	Used to discount this externality (§A.2).
$C_{\text{comp,emissions,nominal}}$	Nominal lifetime loss-compensation emissions cost (\$)	Undiscounted sum over lifetime.
$C_{\text{comp,emissions}}$	Loss-compensation emissions cost (PV) (\$)	PV of annual stream from T_{COD} over T_{lifetime} .

Variables

Equations

1. Total energy compensated (annual).

$$TEC = \alpha \times E_{\text{loss}}.$$

2. Energy mix by operating year. For each operating year $\tau = 1, \dots, T_{\text{lifetime}}$, project-path shares $p_j^{\text{proj}}(\tau)$ from initial mix and per-source growth/decay, normalized so

$$\sum_j p_j^{\text{proj}}(\tau) = 1.$$

3. Emissions in operating year τ (by pollutant).

$$E_{k,\tau} = TEC \times \sum_j p_j^{\text{proj}}(\tau) \times I_{j,k}.$$

4. Cost in operating year τ (by pollutant).

$$C_{k,\tau} = E_{k,\tau} \times c_k.$$

5. Total emissions cost in operating year τ .

$$C_\tau = \sum_k C_{k,\tau}.$$

6. Nominal (undiscounted) lifetime cost.

$$C_{\text{comp,emissions,nominal}} = \sum_{\tau=1}^{T_{\text{lifetime}}} C_\tau.$$

7. Present value.

$$C_{\text{comp,emissions}} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{\tau}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

Both C_{energy} and $C_{\text{comp,emissions}}$ originate from E_{loss} (physical transmission losses), but they use different valuations: C_{energy} applies the value of load to the MWh of lost energy; $C_{\text{comp,emissions}}$ applies externality prices to the pollutant emissions from the make-up generation. They measure different economic effects and do not double-count.

B.4.2 Facilitated emissions

This subsection defines $C_{\text{fac,emissions}}^{(s)}$: the absolute social cost of emissions from instance s 's generation mix over the deliverable energy $E_{\text{delivered,annual}}$ (§C.2.1). $C_{\text{fac,emissions}}^P$ is the externality cost the project actually incurs—the social cost of the generation the line facilitates. It enters the emissions cost bucket alongside $C_{\text{comp,emissions}}^P$. The counterfactual $C_{\text{fac,emissions}}^{\text{no}}$ is computed for use by the displacement module (§C.3) but is not itself a cost anyone incurs.

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Non-double-counting with loss compensation. The loss-compensation subsection (§B.4.1) and this subsection operate on **disjoint MWh pools**: E_{loss} (energy dissipated along the line) vs. $E_{\text{delivered,annual}}$ (throughput reaching the load end). Setting $\alpha = 0$ removes the loss-compensation stream without affecting facilitated emissions, and vice versa. No MWh appears in both formulas.

Non-double-counting with delivered-energy market benefit. The same $E_{\text{delivered,annual}}$ carries one market valuation (v_{load} in §C.2.1) and one externality valuation (c_k per kg here). These measure different economic objects: willingness-to-pay for delivered energy vs. social cost of pollutant emissions.

Transparency: value-of-load internalization. If v_{load} already partially reflects emissions costs (e.g. through carbon pricing, cap-and-trade allowance costs, or renewable energy credits), part of the externality may already be captured in the delivered-energy benefit. Users should either (a) use a value of load net of carbon/emissions adders, or (b) adjust c_k to exclude externalities already reflected in the value of load. The default assumption is that v_{load} reflects a wholesale-level willingness to pay that does not fully internalize the societal externalities valued by c_k .

Variable	Meaning / units	Notes
$E_{\text{delivered,annual}}$	Deliverable energy, (MWh/year)	From §C.2.1. Constant over lifetime.
j	Energy source index, (index)	Same as §B.4.1.
k	Pollutant index, (index)	CO ₂ , SO _x , NO _x ; same as §B.4.1.
$p_j^{\text{proj}}(\tau)$	Project-path share of source j , year τ	From <code>energy_source_mix</code> ; same path as loss-comp (§B.4.1).
$p_j^{\text{no}}(\tau)$	Counterfactual (no-line) share of source j , year τ	From <code>counterfactual_energy_source_mix</code> ; used here and by displacement (§C.3).
g_j^{proj}	Per-source annual rate of change, project	From <code>energy_source_mix[j].rate_of_change</code> .
g_j^{no}	Per-source annual rate of change, no-line	From <code>counterfactual_energy_source_mix[j].rate_of_change</code> .
$I_{j,k}$	Emission intensity, (kg/MWh)	Shared with loss-comp; from <code>16_emissions_reductions.yaml</code> .
$EF_k^{(s)}(\tau)$	Implied emission factor, (kg/MWh)	For instance $s \in \{\text{proj}, \text{no}\}$.
$E_{k,\tau}^{(s)}$	Emissions mass under instance s , year τ , (kg)	
$C_{k,\tau}^{(s)}$	Emissions cost under instance s , year τ , (\$)	
$PV_k^{(s)}$	PV emissions cost for pollutant k , instance s , (\$)	
c_k	Societal cost per kg of pollutant k , (\$/kg)	Same as §B.4.1.
r_{social}	Social discount rate, (decimal)	Same as §B.4.1.
T_{COD}	Commercial operation date, (year)	First year of operation.
T_{lifetime}	Project lifetime, (years)	
$C_{\text{fac,emissions}}^{(s)}$	Absolute facilitated emissions cost, instance s , (\$)	$= \sum_k PV_k^{(s)} \cdot C_{\text{fac,emissions}}^P$ enters $C_{\text{emissions}}^P$.

Variables

Equations

1. Scope (deliverable energy). The scoped MWh is the deliverable energy from §C.2.1, constant over the project lifetime:

$$H_{S,\tau} = E_{\text{delivered,annual}}.$$

2. Mix evolution (both instances). For each instance $s \in \{\text{proj}, \text{no}\}$, starting from the common initial shares $p_j^{(s)}(1)$, the same rate-based compound-growth engine used in §B.4.1 produces time paths. Step 1—compound growth/decay:

$$\tilde{p}_j^{(s)}(\tau) = p_j^{(s)}(1) \cdot (1 + g_j^{(s)})^{\tau-1}, \quad \tau = 1, \dots, T_{\text{lifetime}}.$$

Step 2—renormalize each year:

$$p_j^{(s)}(\tau) = \frac{\tilde{p}_j^{(s)}(\tau)}{\sum_{j'} \tilde{p}_{j'}^{(s)}(\tau)}.$$

The project path uses `energy_source_mix`; the no-line path uses `counterfactual_energy_source_mix`.

3. Implied emission factor. For $s \in \{\text{proj}, \text{no}\}$ and pollutant k :

$$EF_k^{(s)}(\tau) = \sum_j p_j^{(s)}(\tau) I_{j,k}.$$

4. Absolute emissions mass per year (by instance). For instance s and pollutant k :

$$E_{k,\tau}^{(s)} = E_{\text{delivered,annual}} \times EF_k^{(s)}(\tau).$$

5. Absolute emissions cost per year.

$$C_{k,\tau}^{(s)} = E_{k,\tau}^{(s)} \times c_k.$$

6. Present value per pollutant (by instance). Discounted at r_{social} with the same index convention as §B.4.1:

$$PV_k^{(s)} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{k,\tau}^{(s)}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

7. Total across pollutants — absolute facilitated emissions cost.

$$C_{\text{fac,emissions}}^{(s)} = \sum_k PV_k^{(s)}.$$

Pollutants: CO_2 , SO_x , NO_x —the same set as §B.4.1. $C_{\text{fac,emissions}}^P$ enters the emissions cost bucket (§D).

Emissions cost aggregate

$$C_{\text{emissions}}^P = C_{\text{comp,emissions}}^P + C_{\text{fac,emissions}}^P.$$

See §B.4.1 for $C_{\text{comp,emissions}}$ and §B.4.2 for $C_{\text{fac,emissions}}$. The displacement avoided cost $C_{\text{displ,emissions}}^P$ (§C.3) is computed and reported but does not enter this bucket or NB.

C Benefits

This section defines the two benefit buckets. **Remedial benefits** (B_{remedial}) are relief of pre-existing system harms: curtailment reduction (§C.1.1) and congestion reduction (§C.1.2). **Enabling benefits** (B_{enabling}) are new productive value the line creates: the benefit of delivered energy (§C.2.1). Effective capacity relief, capacity remaining after curtailment, and relief quantities ($R_{\text{curtailment,annual}}$, $R_{\text{congestion,...}}$) are defined in §A.4; this section only adds benefit valuation (value per MWh γ , annual benefit B , lifetime nominal and real PV). Curtailment is relieved first, then congestion; overlap of congestion and curtailment hours is addressed in §A.4 and used in §C.1.2. Benefits are not part of rate base or AFUDC. The displacement avoided cost $C_{\text{displ,emissions}}^P$ (§C.3) is computed and reported for transparency but does not enter B^P .

C.1 Remedial Benefits (B_{remedial})

Remedial benefits are relief of pre-existing system harms: curtailment reduction and congestion reduction. $B_{\text{remedial}}^P = B_{\text{curtailment,lifetime}}^P + B_{\text{congestion,lifetime}}^P$.

C.1.1 Curtailment reduction benefit

This subsection defines the curtailment reduction benefit, which contributes to B_{remedial} (§D). Curtailment relief ($\Delta C_{\text{effective}}$, $R_{\text{curtailment,annual}}$) is as defined in §A.4; this subsection only defines the benefit (value of that relief). Curtailment is when generators (e.g. renewables) must cut output because the grid cannot deliver it. The curtailment reduction benefit is the value of the curtailment relief (MWh/year) that the project provides, valued at $\gamma_{\text{curtailment}}$ (\$/MWh). Annual benefit is summed over the project lifetime in nominal terms or discounted to present value at real WACC from the commercial operation date.

Variable	Meaning / units	Notes
$\Delta C_{\text{effective}}$	Effective capacity relief, (MW)	From §A.4.
ϕ	Flow factor, (dimensionless [0,1])	Greenfield: fraction of nameplate that effectively relieves the constraint.
C_{new}	New line capacity, (MW)	Greenfield and reconductoring.
C_{old}	Old line capacity, (MW)	Reconductoring only.
$MW_{\text{curtailment}}$	MW of curtailed resources, (MW)	
$H_{\text{curtailment}}$	Hours of curtailment per year, (h/year)	
$\gamma_{\text{curtailment}}$	Value of curtailed energy, (\$/MWh)	
$R_{\text{curtailment,annual}}$	Curtailment relief due to project, (MWh/year)	From §A.4.
$B_{\text{curtailment,annual}}$	Annual benefit from curtailment reduction, (\$/year)	
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date (year index), (years)	First year of operation; benefit stream starts here.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount benefits (§A.2).
$B_{\text{curtailment,lifetime,nominal}}$	Nominal lifetime curtailment benefit (\$)	
$B_{\text{curtailment,lifetime}}$	Real (PV) lifetime curtailment benefit (\$)	

Variables

Equations As in §A.4, $\Delta C_{\text{effective}}$ and $R_{\text{curtailment,annual}}$ are defined there.

1. Annual curtailment benefit.

$$B_{\text{curtailment,annual}} = R_{\text{curtailment,annual}} \times \gamma_{\text{curtailment}}.$$

2. Nominal lifetime curtailment benefit.

$$B_{\text{curtailment,lifetime,nominal}} = B_{\text{curtailment,annual}} \times T_{\text{lifetime}}.$$

3. Real (present value) lifetime curtailment benefit. Level annual benefit from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$B_{\text{curtailment,lifetime}} = \sum_{t=0}^{T_{\text{lifetime}}-1} \frac{B_{\text{curtailment,annual}}}{(1 + r_{\text{WACC,real}})^{T_{\text{COD}}+t}}.$$

Link to Congestion reduction benefit (§C.1.2). $\Delta C_{\text{remaining}}$ is defined in §A.4 and used for congestion relief and benefit in §C.1.2. The same effective relief and curtailment-first allocation ensure benefits are not double-counted.

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C.1.2 Congestion reduction benefit

This subsection defines the congestion reduction benefit, which contributes to B_{remedial} (§D). Congestion relief ($\Delta C_{\text{remaining}}$, θ , $H_{\text{congestion,overlap}}$, $H_{\text{congestion,nonoverlap}}$, $R_{\text{congestion,...}}$) is as in §A.4; this subsection only defines the benefit (value of that relief). Congestion is when the grid is short of capacity and cannot deliver all the power that is wanted, so value is lost (e.g. higher prices, redispatch). The congestion reduction benefit is the value of the congestion relief (MWh/year) that the project provides, valued at $\gamma_{\text{congestion}}$ (\$/MWh). Relief is allocated after curtailment: on hours that overlap with curtailment, congestion relief is capped by $\Delta C_{\text{remaining}}$; on non-overlap hours, the full $\Delta C_{\text{effective}}$ is available (§A.4). Annual benefit is summed over the project lifetime in nominal terms or discounted to present value at real WACC from the commercial operation date.

Variable	Meaning / units	Notes
$\Delta C_{\text{effective}}$	Effective capacity relief, (MW)	From §A.4.
$MW_{\text{curtailment}}$	MW of curtailed resources, (MW)	From §A.4.
$\Delta C_{\text{remaining}}$	Capacity remaining for congestion after curtailment, (MW)	From §A.4; curtailment addressed first.
θ	Overlap of congestion and curtailment hours	0 = no overlap, 1 = full overlap; 0 if only congestion or only curtailment.
$H_{\text{congestion}}$	Hours of congestion per year, (h/year)	From §A.4.
$H_{\text{curtailment}}$	Hours of curtailment per year, (h/year)	
$H_{\text{congestion,overlap}}$	Congestion hours that overlap with curtailment, (h/year)	
$H_{\text{congestion,nonoverlap}}$	Congestion hours that do not overlap with curtailment, (h/year)	
$X_{\text{congestion}}$	MW of congestion (exceedance), (MW)	
$\gamma_{\text{congestion}}$	Value of congestion, (\$/MWh)	
$R_{\text{congestion,overlap,annual}}$	Overlap-hours congestion relief due to project, (MWh/year)	
$R_{\text{congestion,nonoverlap,annual}}$	Non-overlap-hours congestion relief due to project, (MWh/year)	
$R_{\text{congestion,annual}}$	Total congestion relief due to project, (MWh/year)	
$B_{\text{congestion,annual}}$	Annual benefit from congestion reduction, (\$/year)	
T_{lifetime}	Project lifetime, (years)	Sum of overlap and non-overlap.
T_{COD}	Commercial operation date (year index), (years)	
$r_{\text{WACC,real}}$	Real WACC, (decimal)	
$B_{\text{congestion,lifetime,nominal}}$	Nominal lifetime congestion benefit (\$)	
$B_{\text{congestion,lifetime}}$	Real (PV) lifetime congestion benefit (\$)	First year of operation; benefit stream starts here. Used to discount benefits (§A.2).

Variables

Equations Capacity remaining $\Delta C_{\text{remaining}}$, overlap θ , $H_{\text{congestion,overlap}}$, $H_{\text{congestion,nonoverlap}}$, and relief $R_{\text{congestion,overlap,annual}}$, $R_{\text{congestion,nonoverlap,annual}}$, $R_{\text{congestion,annual}}$ are as in §A.4.

1. Annual congestion benefit.

$$B_{\text{congestion,annual}} = R_{\text{congestion,annual}} \times \gamma_{\text{congestion}}.$$

2. Nominal lifetime congestion benefit.

$$B_{\text{congestion,lifetime,nominal}} = B_{\text{congestion,annual}} \times T_{\text{lifetime}}.$$

3. Real (present value) lifetime congestion benefit. Level annual benefit from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$B_{\text{congestion,lifetime}} = \sum_{t=0}^{T_{\text{lifetime}}-1} \frac{B_{\text{congestion,annual}}}{(1 + r_{\text{WACC,real}})^{T_{\text{COD}}+t}}.$$

Link to Curtailment reduction benefit (§C.1.1). Effective relief $\Delta C_{\text{effective}}$ and curtailment-first allocation (yielding $\Delta C_{\text{remaining}}$) are defined in §A.4, used in §C.1.1. The same allocation ensures congestion and curtailment benefits are consistent and not double-counted.

C.2 Enabling Benefits (B_{enabling})

Enabling benefits are new productive value the line creates: the benefit of delivered energy. $B_{\text{enabling}}^P = B_{\text{delivered,lifetime}}^P$.

C.2.1 Benefit of delivered energy

This subsection defines the benefit of delivered energy, which constitutes B_{enabling} (\$D): the societal value of the energy the project enables to be delivered each year. For greenfield, deliverable capacity equals effective capacity as in §A.4, $\Delta C_{\text{effective}} = \phi \times C_{\text{new}}$. For reconductoring, deliverable capacity is the additional capacity the upgrade enables, $C_{\text{new}} - C_{\text{old}}$ (same as $\Delta C_{\text{effective}}$ in §A.4 for reconductoring); this captures the value of additional energy the upgrade enables, including any headroom beyond clearing the constraint. Deliverable energy per year is deliverable capacity $\times$ line utilization $\times$ hours per year (MWh/year), valued at value of load v_{load} (\$/MWh). Level annual benefit from COD, discounted at real WACC. This benefit is not part of rate base or AFUDC and is not double-counted with congestion or curtailment (those value relief of constraints; this values throughput).

Variable	Meaning / units	Notes
$\Delta C_{\text{effective}}$	Effective capacity relief, (MW)	From §A.4. Greenfield: ϕC_{new} ; reconductoring: $C_{\text{new}} - C_{\text{old}}$.
u	Line utilization, (dimensionless)	From §B.2.4. $0 \leq u \leq 1$.
H	Hours per year, (h/year)	e.g. 8760. From §B.2.4.
v_{load}	Value of load, (\$/MWh)	Marginal value of energy to end-use load. From §B.2.5.
$E_{\text{delivered,annual}}$	Deliverable energy per year, (MWh/year)	
$B_{\text{delivered,annual}}$	Annual benefit from delivered energy, (\$/year)	
T_{lifetime}	Project lifetime, (years)	
T_{COD}	Commercial operation date (year index), (years)	First year of operation; benefit stream starts here.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount benefits (§A.2).
$B_{\text{delivered,lifetime,nominal}}$	Nominal lifetime benefit from delivered energy, (\$)	
$B_{\text{delivered,lifetime}}$	Real (PV) lifetime benefit from delivered energy, (\$)	

Variables

Equations

1. Deliverable energy (annual).

$$E_{\text{delivered,annual}} = \Delta C_{\text{effective}} \times u \times H.$$

2. Annual benefit.

$$B_{\text{delivered,annual}} = E_{\text{delivered,annual}} \times v_{\text{load}}.$$

3. Nominal lifetime benefit.

$$B_{\text{delivered,lifetime,nominal}} = B_{\text{delivered,annual}} \times T_{\text{lifetime}}.$$

4. Real (present value) lifetime benefit. Level annual benefit from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$B_{\text{delivered,lifetime}} = \sum_{t=0}^{T_{\text{lifetime}}-1} \frac{B_{\text{delivered,annual}}}{(1 + r_{\text{WACC,real}})^{T_{\text{COD}}+t}}.$$

Link to §A.4 and Thermal loss costs (§B.2.5). $\Delta C_{\text{effective}}$ is defined in §A.4; v_{load} is defined in §B.2.5. Total societal benefits comprise B_{remedial} (congestion + curtailment) and B_{enabling} (delivered energy); see §D for bucket definitions.

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C.3 Displacement avoided emissions

This subsection derives the displacement avoided emissions cost $C_{\text{displ,emissions}}^P$: the difference between the no-line and project absolute facilitated emissions costs from §B.4.2. Transmission projects that facilitate greater enablement of clean generation leading to displacement of fossil fuels produce a measurable reduction in the social cost of emissions; $C_{\text{displ,emissions}}^P$ is the formal quantification of that displacement in social-cost terms. A positive $C_{\text{displ,emissions}}^P$ means the project-path generation mix is cleaner than the no-line counterfactual—an avoided externality cost. $C_{\text{displ,emissions}}^P$ is computed and **reported for transparency** but does **not** enter NB or BCR; the clean-generation incentive appears through $\Delta C_{\text{emissions}}^{P:B}$ in pairwise comparisons, consistent with how every other avoided-cost story is handled (e.g. avoided risk through $\Delta C_{\text{risk}}^{P:B}$). Not part of rate base or AFUDC.

Variables

Variable	Meaning / units	Notes
$C_{\text{fac,emissions}}^P$	Facilitated emissions cost, project, (\$)	From §B.4.2.
$C_{\text{fac,emissions}}^{\text{no}}$	Facilitated emissions cost, no-line, (\$)	From §B.4.2.
$E_{k,\tau}^{(s)}$	Emissions mass under instance s , year τ , (kg)	From §B.4.2.
$E_{k,\tau}^{\text{avoid}}$	Avoided emissions mass in year τ , (kg)	$= E_{k,\tau}^{\text{no}} - E_{k,\tau}^{\text{proj}}$.
c_k	Societal cost per kg of pollutant k , (\$/kg)	Same as §B.4.1.
$C_{k,\tau}^{\text{avoid}}$	Avoided emissions cost in year τ , (\$)	
r_{social}	Social discount rate, (decimal)	Same as §B.4.1.
T_{COD}	Commercial operation date, (year)	
T_{lifetime}	Project lifetime, (years)	
PV_k^{avoid}	PV avoided cost for pollutant k , (\$)	
$C_{\text{displ,emissions}}^P$	Displacement avoided emissions cost, (\$)	$= C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P$. Reported for transparency; not in NB.

Equations

1. Avoided emissions mass per year. For pollutant k , using the per-instance emissions mass from §B.4.2:

$$E_{k,\tau}^{\text{avoid}} = E_{k,\tau}^{\text{no}} - E_{k,\tau}^{\text{proj}} = E_{\text{delivered,annual}} \times (EF_k^{\text{no}}(\tau) - EF_k^{\text{proj}}(\tau)).$$

Positive $E_{k,\tau}^{\text{avoid}}$ means fewer emissions under the project mix than under the no-line counterfactual. Negative values (project mix dirtier) are allowed.

2. Avoided emissions cost per year.

$$C_{k,\tau}^{\text{avoid}} = E_{k,\tau}^{\text{avoid}} \times c_k.$$

3. PV of avoided cost per pollutant.

$$PV_k^{\text{avoid}} = PV_k^{\text{no}} - PV_k^{\text{proj}} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{k,\tau}^{\text{avoid}}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

4. Total displacement avoided emissions cost.

$$C_{\text{displ,emissions}}^P = C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P = \sum_k PV_k^{\text{avoid}}.$$

Positive = cleaner project = avoided externality. $C_{\text{displ,emissions}}^P$ is reported for transparency; it does not enter NB (see §D).

Pairwise corollary. $\Delta C_{\text{displ,emissions}}^{P:B} = C_{\text{displ,emissions}}^P - C_{\text{displ,emissions}}^B = C_{\text{fac,emissions}}^B - C_{\text{fac,emissions}}^P$ (the no-line reference term cancels). See §D for incremental reporting.

D Reporting Framework

This section assembles the component present values defined in the four cost bucket subsections (§B.1–§B.4) into two reporting scalars (scenario-internal net benefit and incremental net benefit vs a stated baseline) and defines the sign conventions used throughout the paper.

Bucket totals and grand cost

The cost modules are organized into four reporting buckets (defined in §B.1.3, §B.2.8, §B.3.2, and §B.4):

Bucket	Equation	Constituent modules
C_{hard}	$C_{\text{build}} + C_{\text{ROW, capital}} + C_{\text{env. mit}}$	§B.1.1, §B.1.2, §B.1.3
C_{soft}	$C_{\text{operational}} + C_{\text{energy}} + C_{\text{delay}}$	§B.2.1–§B.2.3, §B.2.4–§B.2.5, §B.2.6–§B.2.8
C_{risk}	$C_{\text{wf}} + C_{\text{outage}}$	§B.3
$C_{\text{emissions}}$	$C_{\text{comp, emissions}} + C_{\text{fac, emissions}}$	§B.4.1, §B.4.2

Grand total cost for scenario P :

$$C^P = C_{\text{hard}}^P + C_{\text{soft}}^P + C_{\text{risk}}^P + C_{\text{emissions}}^P.$$

$C_{\text{emissions}}$ ($= C_{\text{comp, emissions}} + C_{\text{fac, emissions}}$) includes loss-compensation emissions (§B.4.1) and facilitated emissions (§B.4.2). The displacement avoided cost $C_{\text{displ, emissions}}^P$ (§C.3) is computed and reported but does not enter this cost total or NB.

Benefit buckets and total benefit

The benefit streams from §C are organized into two reporting buckets:

Bucket	Equation	Constituent modules
B_{remedial}	$B_{\text{curtailment, lifetime}} + B_{\text{congestion, lifetime}}$	§C.1.1, §C.1.2
B_{enabling}	$B_{\text{delivered, lifetime}}$	§C.2.1

Total benefit for scenario P :

$$B^P = B_{\text{remedial}}^P + B_{\text{enabling}}^P.$$

B_{remedial} captures relief of pre-existing system harms (binding congestion, curtailed generation). B_{enabling} captures new productive value (delivered energy throughput). The facilitated emissions cost $C_{\text{fac,emissions}}^P$ (§B.4.2) is in the emissions cost bucket. The displacement avoided cost $C_{\text{displ,emissions}}^P$ (§C.3) is computed and reported for transparency but does not enter B^P or NB.

Scenario-internal net benefit

The scenario-internal net benefit answers “how does scenario P look on its own ledger?”:

$$\text{NB}^P = B^P - C^P.$$

Incremental net benefit vs a stated baseline

For pairwise comparisons, let B denote a stated baseline scenario. The incremental cost convention follows the avoided-cost sign rule: positive means scenario P has lower PV cost than baseline B .

Incremental cost (avoided-cost convention).

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$$\Delta C^{P:B} = C^B - C^P \quad (\Delta C^{P:B} > 0 \Rightarrow P \text{ avoids cost relative to } B).$$

Componentwise:

$$\Delta C_{\text{hard}}^{P:B} = C_{\text{hard}}^B - C_{\text{hard}}^P, \quad \Delta C_{\text{soft}}^{P:B} = C_{\text{soft}}^B - C_{\text{soft}}^P, \quad \Delta C_{\text{risk}}^{P:B} = C_{\text{risk}}^B - C_{\text{risk}}^P, \quad \Delta C_{\text{emissions}}^{P:B} = C_{\text{emissions}}^B - C_{\text{emissions}}^P.$$

Incremental benefits.

$$\Delta B_{\text{remedial}}^{P:B} = B_{\text{remedial}}^P - B_{\text{remedial}}^B, \quad \Delta B_{\text{enabling}}^{P:B} = B_{\text{enabling}}^P - B_{\text{enabling}}^B.$$

A positive ΔB means P delivers more benefit than baseline B in that channel.

Incremental net benefit.

$$\Delta \text{NB}^{P:B} = \Delta B^{P:B} + \Delta C^{P:B}.$$

Consistency check: $\Delta \text{NB}^{P:B} = \text{NB}^P - \text{NB}^B$, since $(B^P - B^B) - (C^P - C^B) = \Delta B^{P:B} + (C^B - C^P)$.

Risk and avoided risk. A lower risk cost under P than under B appears as a positive $\Delta C_{\text{risk}}^{P:B}$ in the incremental table. It is not added to B_{remedial} ; the incremental cost row is the avoided-risk story.

Emissions and displacement. The emissions cost bucket has two components: loss-compensation ($C_{\text{comp,emissions}}^P$) and facilitated ($C_{\text{fac,emissions}}^P$). A project with lower line losses generates lower $C_{\text{comp,emissions}}^P$; a project whose facilitated generation mix is cheaper in externality terms has lower $C_{\text{fac,emissions}}^P$. Both contribute positively to $\Delta C_{\text{emissions}}^{P:B}$ when P is cheaper than baseline B , consistent with how avoided risk appears through $\Delta C_{\text{risk}}^{P:B}$. The displacement avoided cost $C_{\text{displ,emissions}}^P$ (§C.3) is computed and reported but does not enter NB; the clean-generation incentive is fully captured by $\Delta C_{\text{emissions}}^{P:B}$.

Benefit-cost ratios

A benefit-cost ratio under perspective π is:

$$\text{BCR}_{\pi} = \frac{B_{\pi}^P}{C_{\pi}^P}.$$

$\text{BCR}_{\pi} > 1$ means benefits exceed costs under that perspective. The following perspectives are defined:

Perspective	Label	Numerator (B_{π})	Denominator (C_{π})	Notes
System	BCR_{sys}	B^P	C^P	Full societal benchmark. Revenue excluded.
Capital-only	BCR_{cap}	B^P	C_{hard}	Capital screening; demonstrates what narrow views miss.
Capital+delay	$\text{BCR}_{\text{cap+del}}$	B^P	$C_{\text{hard}} + C_{\text{delay}}$	Capital and delay exposure.
Excl wildfire	$\text{BCR}_{\text{excl wf}}$	B^P	$C^P - C_{\text{wf}}$	Regions where wildfire risk is inappropriate.
Excl wf+outage	$\text{BCR}_{\text{excl wf+out}}$	B^P	$C^P - C_{\text{wf}} - C_{\text{outage}}$	Excludes all risk sub-items.
Utility	BCR_{util}	R_{PV}	$C_{\text{hard}} + C_{\text{delay}} + C_{\text{operational}}$	Whether the utility recovers out-of-pocket costs.
Ratepayer	BCR_{rp}	B^P	$R_{\text{PV}} + C_{\text{loss}}$	Whether ratepayers receive more value than what they pay.

Numerator discipline. Societal BCRs use B^P in the numerator. Gains from lower risk, emissions, capital, or delay show up as lower C_{π} in the denominator—not as extra benefit lines in the numerator. This is the same principle as the incremental avoided-cost convention: a positive $\Delta C^{P:B}$ means P avoids cost, not that P earns additional benefit. Revenue (R_{PV} , §E) is a transfer between utility and ratepayer; it enters perspective-specific views but not the societal NB.

E Revenue

This section defines revenue to the utility from rate base: a transfer such that benefit to the utility equals cost to ratepayers (it is not a cost or benefit to society). Both perspectives use the same capital spending (same nominal amounts and timing). Utility (regulatory): nominal spending in CWIP is capitalized with AFUDC to COD, giving RB_{nominal} . Society: same spending stream is discounted to base year at real WACC; AFUDC does not appear. Same inputs and time value; different purpose (rate base vs. PV of spending).

Variables

Variable	Meaning / units	Notes
RB_{nominal}	Nominal rate base at COD, (\$)	AFUDC capital at COD. Sum of build, ROW capital, env. mitigation (see §A.3).
C^{cap}	Total capital in rate base at COD, (\$)	$RB_{\text{nominal}} = C^{\text{cap}}$.
$C_{\text{build}}^{\text{cap}}$	Capitalized build cost at COD, (\$)	Nominal spending capitalized to COD via AFUDC.
$C_{\text{ROW}}^{\text{cap}}$	Capitalized ROW capital cost at COD, (\$)	Nominal spending capitalized to COD via AFUDC.
$C_{\text{env mit}}^{\text{cap}}$	Capitalized environmental mitigation cost at COD, (\$)	Nominal spending capitalized to COD via AFUDC.
C_i^{nom}	Nominal spending (pre-AFUDC) for component i , (\$)	Same timing as for AFUDC; $C_i^{\text{cap}} = C_i^{\text{nom}} + \text{AFUDC}_i$.
π	Inflation rate, (decimal)	
RB_{real}	Real rate base in base-year dollars, (\$)	$RB_{\text{real}} = RB_{\text{nominal}} / (1 + \pi)^{T_{\text{COD}}}$.
ρ	Allowed return rate (real), (decimal)	
$R_{\text{annual,real}}$	Annual revenue in real terms, (\$/year)	
T_{lifetime}	Project lifetime, (years)	
$R_{\text{total,real}}$	Total revenue over lifetime, undiscounted, real, (\$)	Sum of real revenue over T_{lifetime} ; no time value.
$r_{\text{WACC,real}}$	Real WACC, (decimal)	Used to discount the revenue stream.
R_{PV}	Present value of revenue, (\$)	Benefit to utility and cost to ratepayers in PV terms.
T_{COD}	Commercial operation date, (year index)	Years from base year to first year of operation.

Equations

1. Rate base (nominal at COD).

$$RB_{\text{nominal}} = C^{\text{cap}} = C_{\text{build}}^{\text{cap}} + C_{\text{ROW}}^{\text{cap}} + C_{\text{env mit}}^{\text{cap}}.$$

Each C_i^{cap} is AFUDC capital at COD: $C_i^{\text{cap}} = C_i^{\text{nom}} + \text{AFUDC}_i$.

2. COD (year index from base year).

$$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1.$$

3. Real rate base (deflate to base year).

$$RB_{\text{real}} = \frac{RB_{\text{nominal}}}{(1 + \pi)^{T_{\text{COD}}}}.$$

4. Real annual revenue (constant in base-year \$/year).

$$R_{\text{annual,real}} = RB_{\text{real}} \times \rho.$$

5. Total revenue over lifetime (real, undiscounted).

$$R_{\text{total,real}} = R_{\text{annual,real}} \times T_{\text{lifetime}}.$$

6. Present value of revenue (benefit to utility / cost to ratepayers). Revenue starts at T_{COD} and runs for T_{lifetime} years; discount at $r_{\text{WACC,real}}$:

$$R_{\text{PV}} = \sum_{t=0}^{T_{\text{lifetime}}-1} \frac{R_{\text{annual,real}}}{(1 + r_{\text{WACC,real}})^{T_{\text{COD}}+t}}.$$

So $B_{\text{utility}} = R_{\text{PV}}$ and $C_{\text{ratepayers}} = B_{\text{utility}} = R_{\text{PV}}$.